# Vector Calculus: A lecture notes (under preparation)



# **Be careful about typos**

## Contents

# 1 Vector Calculus

## 1.1 Limit of Vector Functions

**Definition 1.1.1.** A vector-valued function of a real variable $t \in \mathbb{R}$ is defined as

$$\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k},$$

where $x(t), y(t), z(t)$ are scalar functions defined over $\mathbb{R}$.

**Definition 1.1.2. Limit of a Vector Function-**

The vector function $\mathbf{a}(t)$ is said to tend to a limit $\mathbf{a}_0$ as $t$ tends to $t_0$, if given a positive number $\varepsilon$, however small, there exists a positive number $\delta$ such that

$$\|\mathbf{a}(t) - \mathbf{a}_0\| < \varepsilon \quad \text{when } 0 < |t - t_0| < \delta$$

and we write

$$\lim_{t \to t_0} \mathbf{a}(t) = \mathbf{a}_0$$

**Component-wise Definition** Let

$$\mathbf{a}(t) = (x(t), y(t), z(t)), \quad \mathbf{a}_0 = (L_1, L_2, L_3).$$

Then,

$$\lim_{t \to t_0} \mathbf{a}(t) = \mathbf{a}_0$$

if and only if

$$\lim_{t \to t_0} x(t) = L_1, \quad \lim_{t \to t_0} y(t) = L_2, \quad \lim_{t \to t_0} z(t) = L_3.$$

### 1.1.1 Algebra of Limits

If limits exist, then:

$$\begin{aligned}
\lim_{t \to t_0} (\mathbf{r}_1(t) + \mathbf{r}_2(t)) &= \lim_{t \to t_0} \mathbf{r}_1(t) + \lim \mathbf{r}_2(t), \\
\lim_{t \to t_0} (c\,\mathbf{r}(t)) &= c \lim_{t \to t_0} \mathbf{r}(t), \quad \text{for any scaler c} \\
\lim_{t \to t_0} (\mathbf{r}_1(t) \cdot \mathbf{r}_2(t)) &= \left(\lim_{t \to t_0} \mathbf{r}_1(t)\right) \cdot \left(\lim_{t \to t_0} \mathbf{r}_2(t)\right), \\
\lim_{t \to t_0} (\mathbf{r}_1(t) \times \mathbf{r}_2(t)) &= (\lim_{t \to t_0} \mathbf{r}_1(t)) \times \left(\lim_{t \to t_0} \mathbf{r}_2(t)\right).
\end{aligned}$$

## 1.2 Continuity of Vector Functions

**Definition 1.2.1.** Vector function $\mathbf{a}(t)$ is said to be continuous at $t = t_0$, if given a positive number $\varepsilon$, however small, there exists a positive number $\delta$ such that

$$|\mathbf{a}(t) - \mathbf{a}(t_0)| < \varepsilon \quad \text{when } |t - t_0| < \delta$$

In other words, $\mathbf{a}(t)$ is continuous at $t = t_0$ if

$$\lim_{t \to t_0} \mathbf{a}(t) = \mathbf{a}(t_0)$$

$\mathbf{a}(t_0)$ being the value of $\mathbf{a}(t)$ at $t = t_0$.

If we represent $\mathbf{a}(t)$ in the form

$$\mathbf{a}(t) = a_1(t)\hat{\imath} + a_2(t)\hat{\jmath} + a_3(t)\hat{k}$$

then $\mathbf{a}(t)$ is continuous at $t = t_0$ if and only if its three components are continuous at $t = t_0$.

### 1.2.1 Properties of Continuous Vector Functions

If $\mathbf{r}(t)$ and $\mathbf{s}(t)$ are continuous, then:

- $\mathbf{r}(t) + \mathbf{s}(t)$ is continuous
- $c\mathbf{r}(t)$ is continuous
- $\mathbf{r}(t) \cdot \mathbf{s}(t)$ is continuous
- $\mathbf{r}(t) \times \mathbf{s}(t)$ is continuous
- $\|\mathbf{r}(t)\|$ is continuous

**Geometrical Interpretation:** A continuous vector function represents a smooth curve in space with no breaks or jumps. Standard algebraic operations preserve continuity.

## 1.3 Differentiability

**Definition 1.3.1.** Let $\mathbf{a}(t)$ and $\mathbf{a}(t+\delta t)$ be the values of $\mathbf{a}$ at neighbouring values $t$ and $t+\delta t$ of the scalar variable $t$ and let $\delta\mathbf{a} = \mathbf{a}(t+\delta t) - \mathbf{a}(t)$ be the change in the value of $\mathbf{a}$ corresponding to change $\delta t$ in $t$. Then, the vector function is said to be differentiable at a point $t$, if the limit

$$\frac{d\mathbf{a}}{dt} = \lim_{\delta t \to 0} \frac{\delta\mathbf{a}}{\delta t} = \lim_{\delta t \to 0} \frac{\mathbf{a}(t+\delta t) - \mathbf{a}(t)}{\delta t}$$

exists.

We call this limit, denoted by $d\mathbf{a}/dt$, as the derivative of $\mathbf{a}(t)$ w.r.t. $t$. Obviously, the derivative of a vector w.r.t. a scalar is a vector.

Geometrically, if $\overrightarrow{OP} = \mathbf{a}(t)$, $\overrightarrow{OQ} = \mathbf{a}(t+\delta t)$, then

$$\delta\mathbf{a} = \mathbf{a}(t+\delta t) - \mathbf{a}(t) = \overrightarrow{PQ}.$$

Consequently, the direction of $\dfrac{d\mathbf{a}}{dt}$ is the limiting direction of $\delta\mathbf{a}/\delta t$, i.e. that of $\overrightarrow{PQ}$ when $Q \to P$, i.e. the tangent at $P$ to the curve described by the terminus of $\mathbf{a}$ as $t$ increases.

If we represent $\mathbf{a}(t)$ in component form

$$\mathbf{a}(t) = a_1(t)\hat{\imath} + a_2(t)\hat{\jmath} + a_3(t)\hat{k}$$

then $\mathbf{a}(t)$ is differentiable at a point $t$ if and only if its three components are differentiable at $t$, and then

$$\frac{d\mathbf{a}}{dt} = \frac{da_1}{dt}\hat{\imath} + \frac{da_2}{dt}\hat{\jmath} + \frac{da_3}{dt}\hat{k}$$

### 1.3.1 Properties of the derivative

If $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are differentiable vector functions of a scalar $t$ and $\phi$ is a differentiable scalar function of $t$, then

1. $\dfrac{d}{dt}(\mathbf{a}+\mathbf{b}) = \dfrac{d\mathbf{a}}{dt} + \dfrac{d\mathbf{b}}{dt}$
2. $\dfrac{d}{dt}(\phi\mathbf{a}) = \dfrac{d\phi}{dt}\mathbf{a} + \phi\dfrac{d\mathbf{a}}{dt}$
3. $\dfrac{d}{dt}(\mathbf{a}\cdot\mathbf{b}) = \mathbf{a}\cdot\dfrac{d\mathbf{b}}{dt} + \mathbf{b}\cdot\dfrac{d\mathbf{a}}{dt}$
4. $\dfrac{d}{dt}(\mathbf{a}\times\mathbf{b}) = \dfrac{d\mathbf{a}}{dt}\times\mathbf{b} + \mathbf{a}\times\dfrac{d\mathbf{b}}{dt}$
5. $\dfrac{d}{dt}\{\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c})\} = \dfrac{d\mathbf{a}}{dt}\cdot(\mathbf{b}\times\mathbf{c}) + \mathbf{a}\cdot\left(\dfrac{d\mathbf{b}}{dt}\times\mathbf{c}\right) + \mathbf{a}\cdot\left(\mathbf{b}\times\dfrac{d\mathbf{c}}{dt}\right)$
6. $\dfrac{d}{dt}\{\mathbf{a}\times(\mathbf{b}\times\mathbf{c})\} = \dfrac{d\mathbf{a}}{dt}\times(\mathbf{b}\times\mathbf{c}) + \mathbf{a}\times\left(\dfrac{d\mathbf{b}}{dt}\times\mathbf{c}\right) + \mathbf{a}\times\left(\mathbf{b}\times\dfrac{d\mathbf{c}}{dt}\right)$

#### Proofs

1.

$$\begin{aligned}
\frac{d}{dt}(\mathbf{a}+\mathbf{b}) &= \lim_{\delta t\to 0}\frac{\{\mathbf{a}(t+\delta t)+\mathbf{b}(t+\delta t)\} - \{\mathbf{a}(t)+\mathbf{b}(t)\}}{\delta t}\\
&= \lim_{\delta t\to 0}\frac{\mathbf{a}(t+\delta t)-\mathbf{a}(t)}{\delta t} + \lim_{\delta t\to 0}\frac{\mathbf{b}(t+\delta t)-\mathbf{b}(t)}{\delta t}\\
&= \frac{d\mathbf{a}}{dt} + \frac{d\mathbf{b}}{dt}
\end{aligned}$$

2.

$$\begin{aligned}
\frac{d}{dt}(\phi\mathbf{a}) &= \lim_{\delta t\to 0}\frac{\phi(t+\delta t)\mathbf{a}(t+\delta t) - \phi(t)\mathbf{a}(t)}{\delta t}\\
&= \lim_{\delta t\to 0}\frac{\phi(t+\delta t)\mathbf{a}(t+\delta t) - \phi(t)\mathbf{a}(t+\delta t) + \phi(t)\mathbf{a}(t+\delta t) - \phi(t)\mathbf{a}(t)}{\delta t}\\
&= \lim_{\delta t\to 0}\frac{\phi(t+\delta t)-\phi(t)}{\delta t}\mathbf{a}(t+\delta t) + \lim_{\delta t\to 0}\phi(t)\frac{\mathbf{a}(t+\delta t)-\mathbf{a}(t)}{\delta t}\\
&= \frac{d\phi}{dt}\mathbf{a} + \phi\frac{d\mathbf{a}}{dt}
\end{aligned}$$

3.

$$\begin{aligned}\frac{d}{dt}(\mathbf{a}\cdot\mathbf{b}) &= \lim_{\delta t\to 0}\frac{\mathbf{a}(t+\delta t)\cdot\mathbf{b}(t+\delta t)-\mathbf{a}(t)\cdot\mathbf{b}(t)}{\delta t}\\ &= \lim_{\delta t\to 0}\frac{\mathbf{a}(t+\delta t)\cdot\mathbf{b}(t+\delta t)-\mathbf{a}(t)\cdot\mathbf{b}(t+\delta t)+\mathbf{a}(t)\cdot\mathbf{b}(t+\delta t)-\mathbf{a}(t)\cdot\mathbf{b}(t)}{\delta t}\\ &= \lim_{\delta t\to 0}\frac{\mathbf{a}(t+\delta t)-\mathbf{a}(t)}{\delta t}\cdot\mathbf{b}(t+\delta t)+\lim_{\delta t\to 0}\mathbf{a}(t)\cdot\frac{\mathbf{b}(t+\delta t)-\mathbf{b}(t)}{\delta t}\\ &= \frac{d\mathbf{a}}{dt}\cdot\mathbf{b}+\mathbf{a}\cdot\frac{d\mathbf{b}}{dt}\end{aligned}$$

4.

$$\begin{aligned}\frac{d}{dt}(\mathbf{a}\times\mathbf{b}) &= \lim_{\delta t\to 0}\frac{\mathbf{a}(t+\delta t)\times\mathbf{b}(t+\delta t)-\mathbf{a}(t)\times\mathbf{b}(t)}{\delta t}\\ &= \lim_{\delta t\to 0}\frac{\mathbf{a}(t+\delta t)\times\mathbf{b}(t+\delta t)-\mathbf{a}(t)\times\mathbf{b}(t+\delta t)+\mathbf{a}(t)\times\mathbf{b}(t+\delta t)-\mathbf{a}(t)\times\mathbf{b}(t)}{\delta t}\\ &= \lim_{\delta t\to 0}\frac{\mathbf{a}(t+\delta t)-\mathbf{a}(t)}{\delta t}\times\mathbf{b}(t+\delta t)+\lim_{\delta t\to 0}\mathbf{a}(t)\times\frac{\mathbf{b}(t+\delta t)-\mathbf{b}(t)}{\delta t}\\ &= \frac{d\mathbf{a}}{dt}\times\mathbf{b}+\mathbf{a}\times\frac{d\mathbf{b}}{dt}\end{aligned}$$

5.

$$\begin{aligned}\frac{d}{dt}\{\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c})\} &= \frac{d\mathbf{a}}{dt}\cdot(\mathbf{b}\times\mathbf{c})+\mathbf{a}\cdot\frac{d}{dt}(\mathbf{b}\times\mathbf{c})\\ &= \frac{d\mathbf{a}}{dt}\cdot(\mathbf{b}\times\mathbf{c})+\mathbf{a}\cdot\left(\frac{d\mathbf{b}}{dt}\times\mathbf{c}+\mathbf{b}\times\frac{d\mathbf{c}}{dt}\right)\\ &= \frac{d\mathbf{a}}{dt}\cdot(\mathbf{b}\times\mathbf{c})+\mathbf{a}\cdot\left(\frac{d\mathbf{b}}{dt}\times\mathbf{c}\right)+\mathbf{a}\cdot\left(\mathbf{b}\times\frac{d\mathbf{c}}{dt}\right)\end{aligned}$$

6.

$$\begin{aligned}\frac{d}{dt}\{\mathbf{a}\times(\mathbf{b}\times\mathbf{c})\} &= \frac{d\mathbf{a}}{dt}\times(\mathbf{b}\times\mathbf{c})+\mathbf{a}\times\frac{d}{dt}(\mathbf{b}\times\mathbf{c})\\ &= \frac{d\mathbf{a}}{dt}\times(\mathbf{b}\times\mathbf{c})+\mathbf{a}\times\left(\frac{d\mathbf{b}}{dt}\times\mathbf{c}+\mathbf{b}\times\frac{d\mathbf{c}}{dt}\right)\\ &= \frac{d\mathbf{a}}{dt}\times(\mathbf{b}\times\mathbf{c})+\mathbf{a}\times\left(\frac{d\mathbf{b}}{dt}\times\mathbf{c}\right)+\mathbf{a}\times\left(\mathbf{b}\times\frac{d\mathbf{c}}{dt}\right)\end{aligned}$$

## 1.4 Derivatives of constant vectors

We have

$$\mathbf{a}=a\hat{a},\quad \frac{d\mathbf{a}}{dt}=\frac{da}{dt}\hat{a}+a\frac{d\hat{a}}{dt}$$

**Theorem 1.4.1.** $\mathbf{a}(t)$ *is a constant vector, both in magnitude and direction, if and only if its derivative is a null vector, i.e.*

$$\frac{d\mathbf{a}}{dt}=0$$

*Proof.* If $\mathbf{a}(t)$ is a constant vector, then $a =$ const. and $\hat{a} =$ constant, therefore

$$\frac{da}{dt} = 0 \quad \text{and} \quad \frac{d\hat{a}}{dt} = 0$$

Hence, $\dfrac{d\mathbf{a}}{dt} = 0$.

Conversely, if $\dfrac{d\mathbf{a}}{dt} = 0$, then integrating, we get $\mathbf{a} =$ constant. □

*Note:* A constant vector is generally denoted by $\mathbf{c}$.

**Theorem 1.4.2.** *If $\hat{a}(t)$ is a unit vector, then*

$$\hat{a} \cdot \frac{d\hat{a}}{dt} = 0 \tag{1}$$

*Proof.* If $\hat{a}(t)$ is a unit vector, then $\hat{a} \cdot \hat{a} = 1$. Differentiating w.r.t. $t$, we get

$$\hat{a} \cdot \frac{d\hat{a}}{dt} + \frac{d\hat{a}}{dt} \cdot \hat{a} = 0 \quad \Rightarrow \quad 2\,\hat{a} \cdot \frac{d\hat{a}}{dt} = 0$$

or

$$\hat{a} \cdot \frac{d\hat{a}}{dt} = 0.$$

□

*Note:* (1) implies that either $\hat{a} = 0$ or $d\hat{a}/dt = 0$ or $d\hat{a}/dt$ is perpendicular to $\hat{a}$. Since $\hat{a} \neq 0$, we conclude that the derivative of a unit vector function $\hat{a}(t)$ is perpendicular to $\hat{a}(t)$, provided it is not the null vector.

**Theorem 1.4.3.** $\mathbf{a}(t)$ *is a vector of constant magnitude if and only if*

$$\mathbf{a} \cdot \frac{d\mathbf{a}}{dt} = 0 \tag{2}$$

*Proof.* We have

$$\begin{aligned}\mathbf{a} \cdot \frac{d\mathbf{a}}{dt} &= a\hat{a} \cdot \left(\hat{a}\frac{da}{dt} + a\frac{d\hat{a}}{dt}\right) \\ &= a^2\hat{a} \cdot \frac{d\hat{a}}{dt} + a\frac{da}{dt}.\end{aligned}$$

Using (1), this reduces to

$$\mathbf{a} \cdot \frac{d\mathbf{a}}{dt} = a\frac{da}{dt}. \tag{3}$$

If $\mathbf{a}(t)$ is a vector of constant magnitude, then $a =$ const., hence $da/dt = 0$. Therefore, from (3),

$$\mathbf{a} \cdot \frac{d\mathbf{a}}{dt} = 0.$$

Conversely, if $\mathbf{a} \cdot \dfrac{d\mathbf{a}}{dt} = 0$, then from (3), $a\,da/dt = 0$, so $da/dt = 0$. Hence $a =$ const. □

*Note:* (2) implies that the derivative of a vector function of constant magnitude is perpendicular to $\mathbf{a}(t)$, provided it is not the null vector.

**Theorem 1.4.4.** $\mathbf{a}(t)$ *is a vector of constant direction if and only if*

$$\mathbf{a} \times \frac{d\mathbf{a}}{dt} = 0$$

*Proof.* We have

$$\begin{aligned} \mathbf{a} \times \frac{d\mathbf{a}}{dt} &= (a\hat{a}) \times \left(\hat{a}\frac{da}{dt} + a\frac{d\hat{a}}{dt}\right) \\ &= a^2 \hat{a} \times \frac{d\hat{a}}{dt}. \end{aligned} \tag{4}$$

If $\mathbf{a}$ has constant direction, then $\hat{a}$ is constant, so $d\hat{a}/dt = 0$. Hence,

$$\mathbf{a} \times \frac{d\mathbf{a}}{dt} = 0.$$

Conversely, if $\mathbf{a} \times \frac{d\mathbf{a}}{dt} = 0$, then from (4),

$$\hat{a} \times \frac{d\hat{a}}{dt} = 0,$$

which implies $\frac{d\hat{a}}{dt}$ is collinear with $\hat{a}$. But since $\hat{a} \cdot \frac{d\hat{a}}{dt} = 0$, this is possible only if $\frac{d\hat{a}}{dt} = 0$. Hence $\hat{a}$ is constant and $\mathbf{a}$ has constant direction.

□

*Note:* The derivative of a vector function of constant direction is collinear with $\mathbf{a}(t)$, provided it is not the null vector.

## 1.5 Radial and Transversal Components

$$\mathbf{r} = \overrightarrow{OP} = r\hat{r}$$

and

$$\hat{r} = \cos\theta\,\hat{i} + \sin\theta\,\hat{j}, \qquad \hat{\theta} = -\sin\theta\,\hat{i} + \cos\theta\,\hat{j}.$$

Consequently,

$$\frac{d\hat{r}}{dt} = \frac{d\theta}{dt}\hat{\theta}, \qquad \frac{d\hat{\theta}}{dt} = -\frac{d\theta}{dt}\hat{r}.$$

### 1.5.1 Velocity

$$\begin{aligned} \mathbf{v} = \frac{d\mathbf{r}}{dt} &= \frac{d}{dt}(r\hat{r}) = \frac{dr}{dt}\hat{r} + r\frac{d\hat{r}}{dt} \\ &= \frac{dr}{dt}\hat{r} + r\frac{d\theta}{dt}\hat{\theta}. \end{aligned} \tag{5}$$

### 1.5.2 Acceleration

$$
\begin{aligned}
\mathbf{a} &= \frac{d\mathbf{v}}{dt} \\
&= \frac{d^2r}{dt^2}\hat{r} + \frac{dr}{dt}\frac{d\hat{r}}{dt} + \frac{dr}{dt}\frac{d\theta}{dt}\hat{\theta} + r\frac{d^2\theta}{dt^2}\hat{\theta} + r\frac{d\theta}{dt}\frac{d\hat{\theta}}{dt} \\
&= \left(\frac{d^2r}{dt^2} - r\left(\frac{d\theta}{dt}\right)^2\right)\hat{r} + \left(2\frac{dr}{dt}\frac{d\theta}{dt} + r\frac{d^2\theta}{dt^2}\right)\hat{\theta}. \qquad (6)
\end{aligned}
$$

The expressions (5) and (6) show that radial velocity is $dr/dt$, transversal velocity is $r\,d\theta/dt$, the radial acceleration is $d^2r/dt^2 - r(d\theta/dt)^2$, and the transversal acceleration is $2(dr/dt)(d\theta/dt) + r\,d^2\theta/dt^2$.

## 1.6 Tangential and Normal Components

Let $P$ be the position of the particle at time $t$, $s$ the arc length of the path from a fixed point $P_0$ to $P$, $\psi$ the angle which the tangent at $P$ makes with the $x$-axis, and $\hat{t}$ and $\hat{n}$ be the unit vectors along tangential and normal directions. Then

$$
\hat{t} = \cos\psi\,\hat{\imath} + \sin\psi\,\hat{\jmath}, \qquad \hat{n} = -\sin\psi\,\hat{\imath} + \cos\psi\,\hat{\jmath}.
$$

Thus,

$$
\frac{d\hat{t}}{dt} = \frac{d\psi}{dt}\hat{n}, \qquad \frac{d\hat{n}}{dt} = -\frac{d\psi}{dt}\hat{t}.
$$

Let $Q$ be the position of the particle at time $t+\delta t$ such that $\overrightarrow{OP} = \mathbf{r}$, $\overrightarrow{OQ} = \mathbf{r} + \delta\mathbf{r}$, and arc $PQ = \delta s$. Then the magnitude of velocity is

$$
v = \left|\frac{d\mathbf{r}}{dt}\right| = \lim_{\delta t\to 0}\frac{|\delta\mathbf{r}|}{\delta t} = \lim_{\delta t\to 0}\frac{\text{chord } PQ}{\delta t} = \frac{ds}{dt}.
$$

Since the velocity is tangential to the path at $P$,

$$
\mathbf{v} = v\hat{t} = \frac{ds}{dt}\hat{t}.
$$

### 1.6.1 Acceleration in Tangential-Normal Form

$$
\begin{aligned}
\mathbf{a} = \frac{d\mathbf{v}}{dt} &= \frac{dv}{dt}\hat{t} + v\frac{d\hat{t}}{dt} \\
&= \frac{dv}{dt}\hat{t} + v\frac{d\psi}{dt}\hat{n}.
\end{aligned}
$$

Since $\dfrac{d\psi}{ds} = \dfrac{1}{\rho}$, where $\rho$ is the radius of curvature, we get

$$
\mathbf{a} = \frac{dv}{dt}\hat{t} + \frac{v^2}{\rho}\hat{n}. \qquad (7)
$$

Thus, tangential velocity is $v = ds/dt$, normal velocity is 0, tangential acceleration is $dv/dt$, and normal acceleration is $v^2/\rho$.

## 1.7 Exercises

1. If $\hat{a}(t)$ is a unit vector, then prove that

   (a) $\hat{a} \cdot \dfrac{d\hat{a}}{dt} = 0$

   (b) $\left|\hat{a} \times \dfrac{d\hat{a}}{dt}\right| = \left|\dfrac{d\hat{a}}{dt}\right|$

2. If $\mathbf{r}(s)$ be the position vector of a point $P$ on a curve $C$, and $s$, the arc length measured from a fixed point on $C$, show that

$$\frac{d\mathbf{r}}{ds} = \hat{t}$$

   where $\hat{t}$ is the unit tangent vector to $C$ at $P$.

3. If $\mathbf{a}(t)$ is a vector function of a scalar $t$, then prove that

   (a) $\mathbf{a} \cdot \dfrac{d\mathbf{a}}{dt} = a\dfrac{da}{dt}$

   (b) $a^3 \dfrac{d}{dt}\left(\dfrac{1}{a}\right) = -\mathbf{a} \cdot \dfrac{d\mathbf{a}}{dt}$

   (c) $a^3 \dfrac{d}{dt}\left(\dfrac{\mathbf{a}}{a}\right) = -\mathbf{a} \times \left(\mathbf{a} \times \dfrac{d\mathbf{a}}{dt}\right)$

   (d) $\dfrac{d}{dt}\left(\mathbf{a} \times \dfrac{d\mathbf{a}}{dt}\right) = \mathbf{a} \times \dfrac{d^2\mathbf{a}}{dt^2}$

   (e) $d\dfrac{d}{dt}\left(\mathbf{a} \cdot \dfrac{d^2\mathbf{a}}{dt^2}\right) = \dfrac{d\mathbf{a}}{dt} \cdot \dfrac{d^2\mathbf{a}}{dt^2} + \mathbf{a} \cdot \dfrac{d^3\mathbf{a}}{dt^3}$

4. If $\mathbf{a}(t)$ and $\mathbf{b}(t)$ are differentiable functions of $t$, then show that

   (a) $\dfrac{d}{dt}\left(\mathbf{a} \cdot \dfrac{d\mathbf{b}}{dt} - \dfrac{d\mathbf{a}}{dt} \cdot \mathbf{b}\right) = \mathbf{a} \cdot \dfrac{d^2\mathbf{b}}{dt^2} - \dfrac{d^2\mathbf{a}}{dt^2} \cdot \mathbf{b}$

   (b) $\dfrac{d}{dt}\left(\mathbf{a} \times \dfrac{d\mathbf{b}}{dt} - \dfrac{d\mathbf{a}}{dt} \times \mathbf{b}\right) = \mathbf{a} \times \dfrac{d^2\mathbf{b}}{dt^2} - \dfrac{d^2\mathbf{a}}{dt^2} \times \mathbf{b}$

# 2 Gradient, Divergence and Curl

## 2.1 Directional derivative

Let $\phi(P)$ be a scalar point function, $C$ a smooth curve through $P$ with unit tangent $\hat{\mathbf{a}}$ at $P$. So that the change in the value of $\phi$ from $P$ to $Q$ is $\delta\phi$. Then,

$$\lim_{\delta s\to 0}\frac{\delta\phi}{\delta s}=\lim_{\delta s\to 0}\frac{\phi(Q)-\phi(P)}{\delta s}.$$

This limit exists and is called the directional derivative of $\phi$ at $P$ in the direction of $\hat{\mathbf{a}}$, and is generally denoted by $\dfrac{\partial\phi}{\partial s}$ or $\dfrac{d\phi}{ds}$.

Since $\dfrac{\partial\phi}{\partial s}$ is the average rate of change of $\phi$ per unit distance in the direction $P$ to $Q$, the directional derivative represents the rate of change of $\phi$ in the direction of $\hat{\mathbf{a}}$.

At a point $P$, there are infinitely many directional derivatives corresponding to different directions. In particular, along the coordinate axes,

$$\frac{\partial\phi}{\partial x},\quad\frac{\partial\phi}{\partial y},\quad\frac{\partial\phi}{\partial z}.$$

Using this, we can write:

$$\frac{d\phi}{ds}=\frac{\partial\phi}{\partial x}\frac{dx}{ds}+\frac{\partial\phi}{\partial y}\frac{dy}{ds}+\frac{\partial\phi}{\partial z}\frac{dz}{ds}. \tag{8}$$

Let the position vectors of $P(x,y,z)$ and $Q(x+\delta x,y+\delta y,z+\delta z)$ be $\mathbf{r}$ and $\mathbf{r}+\delta\mathbf{r}$ respectively:

$$\mathbf{r}=x\hat{\mathbf{i}}+y\hat{\mathbf{j}}+z\hat{\mathbf{k}},\quad \delta\mathbf{r}=\delta x\hat{\mathbf{i}}+\delta y\hat{\mathbf{j}}+\delta z\hat{\mathbf{k}},\quad |\delta\mathbf{r}|=\delta s.$$

Then the unit tangent vector is:

$$\hat{\mathbf{a}}=\frac{d\mathbf{r}}{ds}=\frac{dx}{ds}\hat{\mathbf{i}}+\frac{dy}{ds}\hat{\mathbf{j}}+\frac{dz}{ds}\hat{\mathbf{k}}. \tag{9}$$

Now define the differential operator $\nabla$ (nabla or del):

$$\nabla=\hat{\mathbf{i}}\frac{\partial}{\partial x}+\hat{\mathbf{j}}\frac{\partial}{\partial y}+\hat{\mathbf{k}}\frac{\partial}{\partial z}. \tag{10}$$

Then,

$$\nabla\phi=\hat{\mathbf{i}}\frac{\partial\phi}{\partial x}+\hat{\mathbf{j}}\frac{\partial\phi}{\partial y}+\hat{\mathbf{k}}\frac{\partial\phi}{\partial z}.$$

Using (8), (9), and (10), the directional derivative becomes:

$$\begin{aligned}\frac{\partial\phi}{\partial s}&=\left(\frac{dx}{ds}\hat{\mathbf{i}}+\frac{dy}{ds}\hat{\mathbf{j}}+\frac{dz}{ds}\hat{\mathbf{k}}\right)\cdot\left(\hat{\mathbf{i}}\frac{\partial\phi}{\partial x}+\hat{\mathbf{j}}\frac{\partial\phi}{\partial y}+\hat{\mathbf{k}}\frac{\partial\phi}{\partial z}\right)\\&=\hat{\mathbf{a}}\cdot\nabla\phi.\end{aligned}$$

## 2.2 Gradient

**Definition 2.2.1.** The vector $\nabla\phi$ is called the gradient of the scalar function $\phi$, denoted by grad $\phi$:

$$\text{grad } \phi = \nabla\phi = \hat{\mathbf{i}}\frac{\partial\phi}{\partial x} + \hat{\mathbf{j}}\frac{\partial\phi}{\partial y} + \hat{\mathbf{k}}\frac{\partial\phi}{\partial z}.$$

Thus, $\nabla\phi$ is a vector whose components are the partial derivatives of $\phi$, and it is independent of the coordinate system.

### 2.2.1 Geometrical Interpretation

Let $\phi(x, y, z)$ be continuous and differentiable. Consider two level surfaces:

$$\phi = \text{constant}, \quad \phi + \delta\phi = \text{constant}.$$

Let $P$ and $Q$ be points on these surfaces. Let $\hat{\mathbf{n}}$ be the unit normal at $P$, and $\hat{\mathbf{a}}$ be any direction.

Let

$$PN = \delta n\, \hat{\mathbf{n}}, \quad PQ = \delta s\, \hat{\mathbf{a}}.$$

Then,

$$\frac{\partial\phi}{\partial s} = \lim_{\delta s\to 0}\frac{\delta\phi}{\delta s} = \lim_{\delta n\to 0}\frac{\delta\phi}{\delta n}\cos\theta = \frac{\partial\phi}{\partial n}\cos\theta,$$

or,

$$\frac{\partial\phi}{\partial s} = \frac{\partial\phi}{\partial n}(\hat{\mathbf{n}}\cdot\hat{\mathbf{a}}). \tag{11}$$

Since $\cos\theta \leq 1$, the maximum value of the directional derivative is

$$\left(\frac{\partial\phi}{\partial s}\right)_{\max} = \frac{\partial\phi}{\partial n} \quad \text{when } \theta = 0.$$

Consequently, the vector $\left(\frac{\partial\phi}{\partial n}\right)\hat{n}$ gives the maximum rate of change of $\phi$ in the direction normal to the surface at $P$. This vector is called the **gradient** or slope of the scalar function $\phi$ at $P$.

Hence, we define the gradient of a scalar function $\phi$ as

$$\text{grad } \phi = \frac{\partial\phi}{\partial n}\hat{n}$$

Since we have not used a particular coordinate system, grad $\phi$ is independent of the coordinate system. Further, it has the geometrical meaning that grad $\phi$ at $P$, if nonzero, is normal to the level surface $\phi(x, y, z) = c$ at $P$ and gives the maximum rate of change of $\phi$ at $P$.

### 2.2.2 Equivalence of Two Definitions

By (11), we have

$$\frac{\partial \phi}{\partial x} = \frac{\partial \phi}{\partial n}\hat{n} \cdot \hat{i}, \quad \frac{\partial \phi}{\partial y} = \frac{\partial \phi}{\partial n}\hat{n} \cdot \hat{j}, \quad \frac{\partial \phi}{\partial z} = \frac{\partial \phi}{\partial n}\hat{n} \cdot \hat{k}$$

Hence,

$$\begin{aligned} \frac{\partial \phi}{\partial n}\hat{n} &= \left(\frac{\partial \phi}{\partial n}\hat{n} \cdot \hat{i}\right)\hat{i} + \left(\frac{\partial \phi}{\partial n}\hat{n} \cdot \hat{j}\right)\hat{j} + \left(\frac{\partial \phi}{\partial n}\hat{n} \cdot \hat{k}\right)\hat{k} \\ &= \frac{\partial \phi}{\partial x}\hat{i} + \frac{\partial \phi}{\partial y}\hat{j} + \frac{\partial \phi}{\partial z}\hat{k} = \nabla \phi \end{aligned}$$

Hence, the two definitions are equivalent.

### 2.2.3 Special Gradients

1. **Gradient of $f(r)$:** Let $\vec{OP} = \vec{r}$ be the position vector of point $P(x, y, z)$, so that

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}, \quad r = |\vec{r}| = \sqrt{x^2 + y^2 + z^2}$$

Then,

$$\frac{\partial r}{\partial x} = \frac{x}{r}, \quad \frac{\partial r}{\partial y} = \frac{y}{r}, \quad \frac{\partial r}{\partial z} = \frac{z}{r}$$

Thus,

$$\text{grad } f(r) = \nabla f(r) = \frac{df}{dr}\nabla r = \frac{f'(r)}{r}(x\hat{i} + y\hat{j} + z\hat{k}) = f'(r)\hat{r}$$

2. **Some Particular Cases**

$$\begin{aligned} \text{grad } r^m &= m r^{m-1}\hat{r} \\ \text{grad } r &= \hat{r} = \frac{\vec{r}}{r} \\ \text{grad}\left(\frac{1}{r}\right) &= -\frac{\vec{r}}{r^3} \end{aligned}$$

## 2.3 Divergence of a Vector Field

**Definition 2.3.1.**

$$\text{div } \mathbf{a} = \nabla \cdot \mathbf{a}$$

If $a_1, a_2, a_3$ are the components of $\mathbf{a}$, i.e.,

$$\mathbf{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k},$$

then

$$\begin{aligned} \text{div } \mathbf{a} &= \left(\hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}\right) \cdot (a_1\hat{i} + a_2\hat{j} + a_3\hat{k}) \\ &= \frac{\partial a_1}{\partial x} + \frac{\partial a_2}{\partial y} + \frac{\partial a_3}{\partial z} = \sum \frac{\partial a_i}{\partial x_i} \end{aligned}$$

Clearly, div $\mathbf{a}$ is a scalar function and is independent of the choice of coordinate system.

### 2.3.1 Geometrical Characterization of Divergence

We define the divergence of a differentiable vector function **a** by

$$\text{div } \mathbf{a} = \lim_{V \to 0} \frac{1}{V} \iint_S \mathbf{a} \cdot \hat{n} \, dS$$

i.e., divergence is the limit of the average outward flux per unit volume as the volume shrinks to a point.

### 2.3.2 Physical Meaning of Divergence

Consider a fluid with velocity field $\mathbf{v}(x, y, z)$ having components

$$\mathbf{v} = (v_1, v_2, v_3).$$

Take a small rectangular parallelepiped with sides $\delta x$, $\delta y$, $\delta z$.

Flow in $x$-direction: Velocity at face $ABCD$ is $v_1$. At the opposite face $EFGH$, velocity increases by

$$v_1 + \frac{\partial v_1}{\partial x} \delta x.$$

Flow through face $ABCD$:

$$v_1 \, \delta y \delta z$$

Flow through face $EFGH$:

$$\left( v_1 + \frac{\partial v_1}{\partial x} \delta x \right) \delta y \delta z$$

Net outward flow in $x$-direction:

$$\frac{\partial v_1}{\partial x} \delta x \, \delta y \, \delta z = \frac{\partial v_1}{\partial x} \delta V$$

Similarly, net outward flow in $y$-direction:

$$\frac{\partial v_2}{\partial y} \delta V$$

Net outward flow in $z$-direction:

$$\frac{\partial v_3}{\partial z} \delta V$$

Total Outward Flow

$$\left( \frac{\partial v_1}{\partial x} + \frac{\partial v_2}{\partial y} + \frac{\partial v_3}{\partial z} \right) \delta V$$

Hence, rate of outward flow per unit volume:

$$\frac{\partial v_1}{\partial x} + \frac{\partial v_2}{\partial y} + \frac{\partial v_3}{\partial z} = \text{div } \mathbf{v}$$

**Conclusion:** Thus, divergence of velocity represents the **rate of outward flow of fluid per unit volume**.

### 2.3.3 Special Divergences

1. **Divergence of r:** If $\mathbf{r} = (x, y, z)$ is the position vector of a point in space, then

$$\text{div } \mathbf{r} = \frac{\partial x}{\partial x} + \frac{\partial y}{\partial y} + \frac{\partial z}{\partial z} = 1 + 1 + 1 = 3$$

2. **Notes**

   - Divergence is defined only for vector functions and is a scalar function.
   - div $\mathbf{a}$ is also called the divergence of the vector field defined by $\mathbf{a}$.
   - $\mathbf{a} \cdot \nabla = a_1 \frac{\partial}{\partial x} + a_2 \frac{\partial}{\partial y} + a_3 \frac{\partial}{\partial z}$ is an operator, whereas

   $$\frac{\partial a_1}{\partial x} + \frac{\partial a_2}{\partial y} + \frac{\partial a_3}{\partial z}$$

   is a scalar.

**Definition 2.3.2. Solenoidal Vector:** A vector point function defined and differentiable in a region, whose divergence is zero, is said to be **solenoidal** in that region. Thus,

$$\text{div } \mathbf{a} = 0$$

**Example 2.1** Magnetic field $\mathbf{H}$ is solenoidal, i.e.,

$$\text{div } \mathbf{H} = 0.$$

## 2.4 Curl of a Vector Field

**Definition 2.4.1.** Let $\mathbf{a}(x, y, z)$ be a differentiable vector function. The curl (or rotation) of $\mathbf{a}$ is defined as

$$\text{curl } \mathbf{a} = \nabla \times \mathbf{a} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ a_1 & a_2 & a_3 \end{vmatrix}$$

Expanding,

$$\text{curl } \mathbf{a} = \left( \frac{\partial a_3}{\partial y} - \frac{\partial a_2}{\partial z} \right) \hat{i} + \left( \frac{\partial a_1}{\partial z} - \frac{\partial a_3}{\partial x} \right) \hat{j} + \left( \frac{\partial a_2}{\partial x} - \frac{\partial a_1}{\partial y} \right) \hat{k}$$

Thus, curl $\mathbf{a}$ is a vector function and is independent of the coordinate system.

### 2.4.1 Geometrical Characterization of Curl

Consider a rigid body rotating about a fixed axis with angular velocity $\boldsymbol{\omega}$.

Let

$$\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$$

be the position vector of a point $P$. Then the velocity is

$$\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}.$$

For rotation about the $z$-axis,

$$\boldsymbol{\omega} = \omega \mathbf{k},$$

so

$$\mathbf{v} = \omega \mathbf{k} \times (x\mathbf{i} + y\mathbf{j} + z\mathbf{k}) = -\omega y \mathbf{i} + \omega x \mathbf{j}.$$

Now,

$$\nabla \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ -\omega y & \omega x & 0 \end{vmatrix} = 2\omega \mathbf{k}.$$

Thus,

$$\operatorname{curl} \mathbf{v} = 2\boldsymbol{\omega}.$$

Hence, the curl of velocity has the direction of the axis of rotation and magnitude equal to twice the angular velocity.

### 2.4.2 Special Curls

1. If $\mathbf{r} = (x, y, z)$ is the position vector, then further results follow.

$$\begin{aligned} \textbf{curl } \mathbf{r} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ x & y & z \end{vmatrix} \\ &= \mathbf{i}\left(\frac{\partial z}{\partial y} - \frac{\partial y}{\partial z}\right) + \mathbf{j}\left(\frac{\partial x}{\partial z} - \frac{\partial z}{\partial x}\right) + \mathbf{k}\left(\frac{\partial y}{\partial x} - \frac{\partial x}{\partial y}\right) \\ &= 0\mathbf{i} + 0\mathbf{j} + 0\mathbf{k} = 0 \end{aligned}$$

2. **Note:**

   (a) Curl is defined for vector functions only and is a vector function.

   (b) $\nabla \times \mathbf{a}$ is also called the curl of the vector field defined by $\mathbf{a}$.

   (c)

$$\mathbf{a} \times \nabla \neq \nabla \times \mathbf{a}$$

The left-hand side (LHS) is an operator:

$$\mathbf{i}\left(a_2\frac{\partial}{\partial z} - a_3\frac{\partial}{\partial y}\right) + \mathbf{j}\left(a_3\frac{\partial}{\partial x} - a_1\frac{\partial}{\partial z}\right) + \mathbf{k}\left(a_1\frac{\partial}{\partial y} - a_2\frac{\partial}{\partial x}\right),$$

while the right-hand side (RHS) is a vector:

$$\mathbf{i}\left(\frac{\partial a_3}{\partial y} - \frac{\partial a_2}{\partial z}\right) + \mathbf{j}\left(\frac{\partial a_1}{\partial z} - \frac{\partial a_3}{\partial x}\right) + \mathbf{k}\left(\frac{\partial a_2}{\partial x} - \frac{\partial a_1}{\partial y}\right).$$

**Definition 2.4.2. Irrotational vector**: A vector point function, defined and differentiable in a certain region of space, whose curl vanishes, is said to be irrotational in that region. Thus, a vector function $\mathbf{a}$ is irrotational if

$$\nabla \times \mathbf{a} = 0$$

For example, the position vector $\mathbf{r}$ is irrotational as $\nabla \times \mathbf{r} = 0$.

## 2.5 Invariance

Most physical quantities are invariant and do not depend upon the choice of the coordinate system, i.e., if a point $P$ in space has coordinates $(x, y, z)$ relative to one coordinate system and the same point has coordinates $(x', y', z')$ relative to another coordinate system, and a scalar point function $\phi$ or a vector point function $\mathbf{a}$ is evaluated at $P$, and has value $\phi(x, y, z)$ or $\mathbf{a}(x, y, z)$ at $P$ with coordinates $(x, y, z)$ and value $\phi(x', y', z')$ or $\mathbf{a}'(x', y', z')$ at $P$ with coordinates $(x', y', z')$, and if

$$\phi(x, y, z) = \phi'(x', y', z'), \quad \mathbf{a}(x, y, z) = \mathbf{a}'(x', y', z')$$

then $\phi$ and $\mathbf{a}$ are invariants under coordinate transformations.

For example, the temperature at a point or the velocity of a fluid at a point is not dependent on whether coordinates $(x, y, z)$ or $(x', y', z')$ are used.

**Theorem 2.5.1.** *The gradient of an invariant scalar point function $\phi$ and the divergence and curl of an invariant vector point function* $\mathbf{a}$ *are invariant under the coordinate transformations.*

*Proof.* By hypothesis, $\phi(x, y, z) = \phi'(x', y', z')$ and $\mathbf{a}(x, y, z) = \mathbf{a}'(x', y', z')$; and let $\nabla\phi = \mathbf{V}\phi$, div $\mathbf{a} = \nabla \cdot \mathbf{a}$, curl $\mathbf{a} = \nabla \times \mathbf{a}$; to prove that

$$\nabla\phi = \nabla'\phi', \quad \nabla \cdot \mathbf{a} = \nabla' \cdot \mathbf{a}', \quad \nabla \times \mathbf{a} = \nabla' \times \mathbf{a}'$$

it is sufficient to prove that $\nabla = \nabla'$ where

$$\nabla = \mathbf{i}\frac{\partial}{\partial x} + \mathbf{j}\frac{\partial}{\partial y} + \mathbf{k}\frac{\partial}{\partial z} \quad \text{and} \quad \nabla' = \mathbf{i}'\frac{\partial}{\partial x'} + \mathbf{j}'\frac{\partial}{\partial y'} + \mathbf{k}'\frac{\partial}{\partial z'}.$$

Let the two rectangular coordinate systems $xyz$ and $x'y'z'$ have the same origin $O$, differing only by rotation. Let the direction cosines of the axes $ox'$, $oy'$, $oz'$ wrt $x, y, z$ be $(l_1, m_1, n_1)$, $(l_2, m_2, n_2)$, $(l_3, m_3, n_3)$ respectively. Then the direction cosines of axes $ox, oy, oz$ wrt $x', y', z'$ are $(l_1, l_2, l_3)$, $(m_1, m_2, m_3)$, $(n_1, n_2, n_3)$. Since the two systems are orthogonal,

$$\begin{aligned} l_1 l_1' + m_1 m_1' + n_1 n_1' &= 0, \\ l_1^2 + m_1^2 + n_1^2 &= 1, \text{ etc.} \end{aligned}$$

and

$$l_m l_1' + l_2 m_2' + l_3 m_3' = 0,$$
$$l_1^2 + l_2^2 + l_3^2 = 1, \text{ etc.}$$

and since the components of a unit vector are its direction cosines,

$$\begin{aligned}
\hat{\imath}' &= l_1\hat{\imath} + m_1\hat{\jmath} + n_1\hat{k} \\
\hat{\jmath}' &= l_2\hat{\imath} + m_2\hat{\jmath} + n_2\hat{k} \\
\hat{k}' &= l_3\hat{\imath} + m_3\hat{\jmath} + n_3\hat{k}
\end{aligned}$$

Let $\mathbf{r}$ and $\mathbf{r}'$ be the position vectors of $P$ in the two systems, then

$$\mathbf{r} = x\hat{\imath} + y\hat{\jmath} + z\hat{k}, \quad \mathbf{r}' = x'\hat{\imath}' + y'\hat{\jmath}' + z'\hat{k}'$$

Since $\mathbf{r} = \mathbf{r}'$, we have

$$\begin{aligned}
x\hat{\imath} + y\hat{\jmath} + z\hat{k} &= x'\hat{\imath}' + y'\hat{\jmath}' + z'\hat{k}' \\
&= x'(l_1\hat{\imath} + m_1\hat{\jmath} + n_1\hat{k}) + y'(l_2\hat{\imath} + m_2\hat{\jmath} + n_2\hat{k}) + z'(l_3\hat{\imath} + m_3\hat{\jmath} + n_3\hat{k}) \\
&= (l_1x' + l_2y' + l_3z')\hat{\imath} + (m_1x' + m_2y' + m_3z')\hat{\jmath} + (n_1x' + n_2y' + n_3z')\hat{k}
\end{aligned}$$

Hence, transformation equations are:

$$\begin{cases}
x = l_1x' + l_2y' + l_3z' \\
y = m_1x' + m_2y' + m_3z' \\
z = n_1x' + n_2y' + n_3z'
\end{cases}$$

Now, using chain rule:

$$\frac{\partial}{\partial x'} = \frac{\partial x}{\partial x'}\frac{\partial}{\partial x} + \frac{\partial y}{\partial x'}\frac{\partial}{\partial y} + \frac{\partial z}{\partial x'}\frac{\partial}{\partial z}$$

Thus,

$$\frac{\partial}{\partial x'} = l_1\frac{\partial}{\partial x} + m_1\frac{\partial}{\partial y} + n_1\frac{\partial}{\partial z}$$

Similarly,

$$\begin{aligned}
\frac{\partial}{\partial y'} &= l_2\frac{\partial}{\partial x} + m_2\frac{\partial}{\partial y} + n_2\frac{\partial}{\partial z} \\
\frac{\partial}{\partial z'} &= l_3\frac{\partial}{\partial x} + m_3\frac{\partial}{\partial y} + n_3\frac{\partial}{\partial z}
\end{aligned}$$

Hence,

$$\begin{aligned}
\nabla' &= \hat{\imath}'\frac{\partial}{\partial x'} + \hat{\jmath}'\frac{\partial}{\partial y'} + \hat{k}'\frac{\partial}{\partial z'} \\
&= (l_1\hat{\imath} + m_1\hat{\jmath} + n_1\hat{k})\left(l_1\frac{\partial}{\partial x} + m_1\frac{\partial}{\partial y} + n_1\frac{\partial}{\partial z}\right) + (l_2\hat{\imath} + m_2\hat{\jmath} + n_2\hat{k})\left(l_2\frac{\partial}{\partial x} + m_2\frac{\partial}{\partial y} + n_2\frac{\partial}{\partial z}\right) \\
&\quad + (l_3\hat{\imath} + m_3\hat{\jmath} + n_3\hat{k})\left(l_3\frac{\partial}{\partial x} + m_3\frac{\partial}{\partial y} + n_3\frac{\partial}{\partial z}\right)
\end{aligned}$$

Substituting, we obtain

$$\nabla' = \hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z} = \nabla$$

$\square$

The invariance property of gradient, divergence, and curl still holds.

## 2.6 Derivations of Vector Identities

Let $\phi(x, y, z)$ and $\psi(x, y, z$ be differentiable scalar functions and let $\mathbf{a}(x, y, z)$, $\mathbf{b}(x, y, z)$ be differentiable vector functions in space. Then we have following vector identities which are also called expansion formulae and are independent of the choice of coordinate system.

1. $\nabla(\phi \pm \psi) = \nabla\phi \pm \nabla\psi$ [grad]
2. $\nabla \cdot (\mathbf{a} \pm \mathbf{b}) = \nabla \cdot \mathbf{a} \pm \nabla \cdot \mathbf{b}$ [div]
3. $\nabla(\mathbf{a} \times \mathbf{b}) = \nabla\mathbf{a} \times \nabla\mathbf{b}$ [curl]
4. $\nabla(\phi\psi) = \phi\nabla\psi + \psi\nabla\phi$
5. $\nabla\left(\dfrac{\phi}{\psi}\right) = \dfrac{\psi\nabla\phi - \phi\nabla\psi}{\psi^2}, \quad \psi \neq 0$
6. $\nabla(\mathbf{a} \cdot \mathbf{b}) = \mathbf{a} \times (\nabla \times \mathbf{b}) + \mathbf{b} \times (\nabla \times \mathbf{a}) + (\mathbf{a} \cdot \nabla)\mathbf{b} + (\mathbf{b} \cdot \nabla)\mathbf{a}$
7. $\nabla \cdot (\phi\mathbf{a}) = \phi\nabla \cdot \mathbf{a} + \mathbf{a} \cdot \nabla\phi$
8. $\nabla \cdot (\mathbf{a} \times \mathbf{b}) = \mathbf{b} \cdot (\nabla \times \mathbf{a}) - \mathbf{a} \cdot (\nabla \times \mathbf{b})$
9. $\nabla \times (\phi\mathbf{a}) = \phi(\nabla \times \mathbf{a}) + (\nabla\phi) \times \mathbf{a}$
10. $\nabla \times (\mathbf{a} \times \mathbf{b}) = \mathbf{a}(\nabla \cdot \mathbf{b}) - \mathbf{b}(\nabla \cdot \mathbf{a}) + (\mathbf{b} \cdot \nabla)\mathbf{a} - (\mathbf{a} \cdot \nabla)\mathbf{b}$

## Proofs

1. Gradient of sum:

$$\begin{aligned}
\nabla(\phi + \psi) &= \sum_i \hat{e}_i \frac{\partial}{\partial x_i}(\phi + \psi) \\
&= \sum_i \hat{e}_i \left(\frac{\partial \phi}{\partial x_i} + \frac{\partial \psi}{\partial x_i}\right) \\
&= \nabla\phi + \nabla\psi
\end{aligned}$$

2. Divergence of sum:

$$\begin{aligned}\nabla \cdot (\mathbf{a}+\mathbf{b}) &= \sum_i \frac{\partial}{\partial x_i}(a_i + b_i) \\ &= \sum_i \left(\frac{\partial a_i}{\partial x_i} + \frac{\partial b_i}{\partial x_i}\right) \\ &= \nabla \cdot \mathbf{a} + \nabla \cdot \mathbf{b}\end{aligned}$$

3. Curl of sum:

$$\begin{aligned}\nabla \times (\mathbf{a}+\mathbf{b}) &= \sum_i \hat{e}_i \times \frac{\partial}{\partial x_i}(\mathbf{a}+\mathbf{b}) \\ &= \nabla \times \mathbf{a} + \nabla \times \mathbf{b}\end{aligned}$$

4. Gradient of product:

$$\begin{aligned}\nabla(\phi\psi) &= \sum_i \hat{e}_i \frac{\partial}{\partial x_i}(\phi\psi) \\ &= \sum_i \hat{e}_i \left(\phi \frac{\partial \psi}{\partial x_i} + \psi \frac{\partial \phi}{\partial x_i}\right) \\ &= \phi \nabla \psi + \psi \nabla \phi\end{aligned}$$

5. Gradient of quotient:

$$\begin{aligned}\nabla\left(\frac{\phi}{\psi}\right) &= \sum_i \hat{e}_i \frac{\partial}{\partial x_i}\left(\frac{\phi}{\psi}\right) \\ &= \sum_i \hat{e}_i \left(\frac{\psi \frac{\partial \phi}{\partial x_i} - \phi \frac{\partial \psi}{\partial x_i}}{\psi^2}\right) \\ &= \frac{\psi \nabla \phi - \phi \nabla \psi}{\psi^2}\end{aligned}$$

6. Gradient of dot product: $\nabla(\mathbf{a} \cdot \mathbf{b})$

$$\begin{aligned}\mathbf{b} \times \text{curl } \mathbf{a} = \mathbf{b} \times (\nabla \times \mathbf{a}) &= \mathbf{b} \times \left(\sum_i \hat{\mathbf{e}}_i \times \frac{\partial \mathbf{a}}{\partial x_i}\right) \\ &= \sum_i \mathbf{b} \times \left(\hat{\mathbf{e}}_i \times \frac{\partial \mathbf{a}}{\partial x_i}\right) = \sum_i \left[\left(\mathbf{b} \cdot \frac{\partial \mathbf{a}}{\partial x_i}\right)\hat{\mathbf{e}}_i - (\mathbf{b} \cdot \hat{\mathbf{e}}_i)\frac{\partial \mathbf{a}}{\partial x_i}\right] \\ &= \sum_i \hat{\mathbf{e}}_i \left(\mathbf{b} \cdot \frac{\partial \mathbf{a}}{\partial x_i}\right) - \sum_i (\mathbf{b} \cdot \hat{\mathbf{e}}_i)\frac{\partial \mathbf{a}}{\partial x_i}\end{aligned}$$

$$\therefore \quad \sum_i \hat{\mathbf{e}}_i \left(\mathbf{b} \cdot \frac{\partial \mathbf{a}}{\partial x_i}\right) = \mathbf{b} \times \text{curl } \mathbf{a} + \sum_i (\mathbf{b} \cdot \hat{\mathbf{e}}_i)\frac{\partial \mathbf{a}}{\partial x_i}$$

Similarly,

$$\sum_i \hat{\mathbf{e}}_i \left(\mathbf{a} \cdot \frac{\partial \mathbf{b}}{\partial x_i}\right) = \mathbf{a} \times \text{curl } \mathbf{b} + \sum_i (\mathbf{a} \cdot \hat{\mathbf{e}}_i) \frac{\partial \mathbf{b}}{\partial x_i}$$

Consequently, we have

$$\begin{aligned}
\nabla(\mathbf{a} \cdot \mathbf{b}) &= \sum_i \hat{\mathbf{e}}_i \frac{\partial}{\partial x_i}(\mathbf{a} \cdot \mathbf{b}) = \sum_i \hat{\mathbf{e}}_i \left(\frac{\partial \mathbf{a}}{\partial x_i} \cdot \mathbf{b} + \mathbf{a} \cdot \frac{\partial \mathbf{b}}{\partial x_i}\right) \\
&= \sum_i \hat{\mathbf{e}}_i \left(\mathbf{b} \cdot \frac{\partial \mathbf{a}}{\partial x_i}\right) + \sum_i \hat{\mathbf{e}}_i \left(\mathbf{a} \cdot \frac{\partial \mathbf{b}}{\partial x_i}\right) \\
&= \mathbf{b} \times \text{curl } \mathbf{a} + \sum_i (\mathbf{b} \cdot \hat{\mathbf{e}}_i) \frac{\partial \mathbf{a}}{\partial x_i} + \mathbf{a} \times \text{curl } \mathbf{b} + \sum_i (\mathbf{a} \cdot \hat{\mathbf{e}}_i) \frac{\partial \mathbf{b}}{\partial x_i}
\end{aligned}$$

$$= \mathbf{a} \times \text{curl } \mathbf{b} + \mathbf{b} \times \text{curl } \mathbf{a} + (\mathbf{a} \cdot \nabla)\mathbf{b} + (\mathbf{b} \cdot \nabla)\mathbf{a}$$

7. Divergence of scalar-vector product:

$$\begin{aligned}
\nabla \cdot (\phi \mathbf{a}) &= \sum_i \frac{\partial}{\partial x_i}(\phi a_i) \\
&= \sum_i \left(\phi \frac{\partial a_i}{\partial x_i} + a_i \frac{\partial \phi}{\partial x_i}\right) \\
&= \phi \nabla \cdot \mathbf{a} + \mathbf{a} \cdot \nabla \phi
\end{aligned}$$

8. Divergence of cross product:

$$\begin{aligned}
\nabla \cdot (\mathbf{a} \times \mathbf{b}) &= \sum \hat{\imath} \cdot \frac{\partial}{\partial x} (\mathbf{a} \times \mathbf{b}) = \sum \hat{\imath} \left(\frac{\partial \mathbf{a}}{\partial x} \times \mathbf{b} + \mathbf{a} \times \frac{\partial \mathbf{b}}{\partial x}\right) \\
&= \sum \left[\hat{\imath}, \frac{\partial \mathbf{a}}{\partial x}, \mathbf{b}\right] + \sum \left[\hat{\imath}, \mathbf{a}, \frac{\partial \mathbf{b}}{\partial x}\right] \\
&= \sum \left[\mathbf{b}, \hat{\imath}, \frac{\partial \mathbf{a}}{\partial x}\right] - \sum \left[\mathbf{a}, \hat{\imath}, \frac{\partial \mathbf{b}}{\partial x}\right] \\
&= \sum \mathbf{b} \cdot \hat{\imath} \times \frac{\partial \mathbf{a}}{\partial x} - \sum \mathbf{a} \cdot \hat{\imath} \times \frac{\partial \mathbf{b}}{\partial x} \\
&= \mathbf{b} \cdot \sum \hat{\imath} \times \frac{\partial \mathbf{a}}{\partial x} - \mathbf{a} \cdot \sum \hat{\imath} \times \frac{\partial \mathbf{b}}{\partial x} \\
&= \mathbf{b} \cdot (\nabla \times \mathbf{a}) - \mathbf{a} \cdot (\nabla \times \mathbf{b})
\end{aligned}$$

9. Curl of scalar-vector product: $\nabla \times (\phi \mathbf{a})$

$$\begin{aligned}
\nabla \times (\phi \mathbf{a}) &= \sum \hat{\imath} \frac{\partial}{\partial x}(\phi \mathbf{a}) \\
&= \sum \hat{\imath} \left(\frac{\partial \phi}{\partial x} \mathbf{a} + \phi \frac{\partial \mathbf{a}}{\partial x}\right) \\
&= \sum \hat{\imath} \frac{\partial \phi}{\partial x} \times \mathbf{a} + \phi \sum \hat{\imath} \times \frac{\partial \mathbf{a}}{\partial x} \\
&= (\nabla \phi) \times \mathbf{a} + \phi (\nabla \times \mathbf{a})
\end{aligned}$$

10. Curl of cross product $\nabla \times (\mathbf{a} \times \mathbf{b})$

$$\begin{aligned}
\nabla \times (\mathbf{a} \times \mathbf{b}) &= \sum \hat{i} \frac{\partial}{\partial x}(\mathbf{a} \times \mathbf{b}) \\
&= \sum \hat{i} \left( \frac{\partial \mathbf{a}}{\partial x} \times \mathbf{b} + \mathbf{a} \times \frac{\partial \mathbf{b}}{\partial x} \right) \\
&= \sum \hat{i} \times \left( \frac{\partial \mathbf{a}}{\partial x} \times \mathbf{b} \right) + \sum \hat{i} \times \left( \mathbf{a} \times \frac{\partial \mathbf{b}}{\partial x} \right)
\end{aligned}$$

Using the vector triple product identity:

$$\mathbf{p} \times (\mathbf{q} \times \mathbf{r}) = \mathbf{q}(\mathbf{p} \cdot \mathbf{r}) - \mathbf{r}(\mathbf{p} \cdot \mathbf{q}),$$

we get:

$$\begin{aligned}
\nabla \times (\mathbf{a} \times \mathbf{b}) &= \sum \left[ \mathbf{b} \left( \hat{i} \cdot \frac{\partial \mathbf{a}}{\partial x} \right) - \frac{\partial \mathbf{a}}{\partial x}(\hat{i} \cdot \mathbf{b}) \right] \\
&\quad + \sum \left[ \frac{\partial \mathbf{b}}{\partial x}(\hat{i} \cdot \mathbf{a}) - \mathbf{a} \left( \hat{i} \cdot \frac{\partial \mathbf{b}}{\partial x} \right) \right] \\
&= \mathbf{b} \sum \left( \hat{i} \cdot \frac{\partial \mathbf{a}}{\partial x} \right) - \sum \left( \hat{i} \cdot \mathbf{b} \right) \frac{\partial \mathbf{a}}{\partial x} \\
&\quad + \sum \left( \hat{i} \cdot \mathbf{a} \right) \frac{\partial \mathbf{b}}{\partial x} - \mathbf{a} \sum \left( \hat{i} \cdot \frac{\partial \mathbf{b}}{\partial x} \right)
\end{aligned}$$

Recognizing the operators:

$$\nabla \times (\mathbf{a} \times \mathbf{b}) = \mathbf{b}(\nabla \cdot \mathbf{a}) - (\mathbf{b} \cdot \nabla)\mathbf{a} + (\mathbf{a} \cdot \nabla)\mathbf{b} - \mathbf{a}(\nabla \cdot \mathbf{b})$$

Rewriting in standard form:

$$\nabla \times (\mathbf{a} \times \mathbf{b}) = (\mathbf{a} \cdot \nabla)\mathbf{b} - (\mathbf{b} \cdot \nabla)\mathbf{a} + \mathbf{b}(\nabla \cdot \mathbf{a}) - \mathbf{a}(\nabla \cdot \mathbf{b})$$

## 2.7 Second order partial derivatives

Let $\phi(x, y, z)$ and $\mathbf{a}(x, y, z)$ be twice differentiable scalar point function and vector point function, respectively. Then, we may obtain their second order partial derivatives such as

$$\frac{\partial^2 \phi}{\partial x^2}, \quad \frac{\partial^2 \phi}{\partial x \partial y}, \quad \ldots, \quad \frac{\partial^2 a}{\partial x^2}, \quad \frac{\partial^2 a}{\partial x \partial y}, \quad \ldots$$

### 2.7.1 Laplace operator

The following operator, denoted by $\nabla^2$ (read as del squared, nabla squared or $\Delta$ (read as delta)), is called Laplace operator or simply *Laplacian* and frequently occurs in Physics and Mechanics:

$$\nabla^2 = \Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}.$$

Thus, if $\phi(x, y, z)$ is a twice differentiable scalar function, then

$$\nabla^2\phi = \frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} + \frac{\partial^2\phi}{\partial z^2}.$$

and if $\phi$ is such that

$$\nabla^2\phi = \frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} + \frac{\partial^2\phi}{\partial z^2} = 0, \tag{12}$$

then the equation (12) is called *Laplace's equation* and $\phi$ is said to be *harmonic*.

The Laplace operator may be applied to a vector function with the meaning

$$\nabla^2\mathbf{a} = \hat{\imath}\nabla^2 a_1 + \hat{\jmath}\nabla^2 a_2 + \hat{k}\nabla^2 a_3,$$

where $\mathbf{a} = a_1\hat{\imath} + a_2\hat{\jmath} + a_3\hat{k}$.

### 2.7.2 Second Order Vector Identities

Let $\phi(x, y, z)$ and $\mathbf{a}(x, y, z)$ be twice differentiable scalar and vector functions in space. Then, $\nabla\phi$, $\nabla\cdot\mathbf{a}$ and $\nabla\times\mathbf{a}$ are vector, scalar and vector functions, respectively. Hence, these may be combined to give rise to the following second-order vector identities which are also called *second-order expansion formulae* and are independent of the choice of coordinate system.

1. $\operatorname{div}\nabla\phi = \nabla^2\phi$
2. $\operatorname{curl}\nabla\phi = 0$
3. $\operatorname{div}\,\operatorname{curl}\mathbf{a} = 0$
4. $\operatorname{curl}\,\operatorname{curl}\mathbf{a} = \nabla(\operatorname{div}\mathbf{a}) - \nabla^2\mathbf{a}$
5. $\nabla(\operatorname{div}\mathbf{a}) = \operatorname{curl}\,\operatorname{curl}\mathbf{a} + \nabla^2\mathbf{a}$

**Proof:**

1.

$$\begin{aligned}\operatorname{div}\nabla\phi = \nabla\cdot\nabla\phi &= \nabla\cdot\left(\hat{\imath}\frac{\partial\phi}{\partial x} + \hat{\jmath}\frac{\partial\phi}{\partial y} + \hat{k}\frac{\partial\phi}{\partial z}\right)\\ &= \frac{\partial}{\partial x}\left(\frac{\partial\phi}{\partial x}\right) + \frac{\partial}{\partial y}\left(\frac{\partial\phi}{\partial y}\right) + \frac{\partial}{\partial z}\left(\frac{\partial\phi}{\partial z}\right)\\ &= \frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} + \frac{\partial^2\phi}{\partial z^2} = \nabla^2\phi\end{aligned}$$

2.

$$\operatorname{curl} \nabla\phi = \nabla \times (\nabla\phi) = \nabla \times \left(\hat{i}\frac{\partial\phi}{\partial x} + \hat{j}\frac{\partial\phi}{\partial y} + \hat{k}\frac{\partial\phi}{\partial z}\right)$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial\phi}{\partial x} & \frac{\partial\phi}{\partial y} & \frac{\partial\phi}{\partial z} \end{vmatrix}$$

$$= \left(\frac{\partial^2\phi}{\partial y\partial z} - \frac{\partial^2\phi}{\partial z\partial y}\right)\hat{i} + \left(\frac{\partial^2\phi}{\partial z\partial x} - \frac{\partial^2\phi}{\partial x\partial z}\right)\hat{j} + \left(\frac{\partial^2\phi}{\partial x\partial y} - \frac{\partial^2\phi}{\partial y\partial x}\right)\hat{k} = 0$$

3. Since

$$\operatorname{curl} \mathbf{a} = \left(\frac{\partial a_3}{\partial y} - \frac{\partial a_2}{\partial z}\right)\hat{i} + \left(\frac{\partial a_1}{\partial z} - \frac{\partial a_3}{\partial x}\right)\hat{j} + \left(\frac{\partial a_2}{\partial x} - \frac{\partial a_1}{\partial y}\right)\hat{k}$$

$$\operatorname{div}\operatorname{curl} \mathbf{a} = \frac{\partial}{\partial x}\left(\frac{\partial a_3}{\partial y} - \frac{\partial a_2}{\partial z}\right) + \frac{\partial}{\partial y}\left(\frac{\partial a_1}{\partial z} - \frac{\partial a_3}{\partial x}\right) + \frac{\partial}{\partial z}\left(\frac{\partial a_2}{\partial x} - \frac{\partial a_1}{\partial y}\right)$$

$$= \frac{\partial^2 a_3}{\partial x\partial y} - \frac{\partial^2 a_2}{\partial x\partial z} + \frac{\partial^2 a_1}{\partial y\partial z} - \frac{\partial^2 a_3}{\partial y\partial x} + \frac{\partial^2 a_2}{\partial z\partial x} - \frac{\partial^2 a_1}{\partial z\partial y} = 0$$

4. Since

$$\operatorname{curl} \mathbf{a} = \left(\frac{\partial a_3}{\partial y} - \frac{\partial a_2}{\partial z}\right)\hat{i} + \left(\frac{\partial a_1}{\partial z} - \frac{\partial a_3}{\partial x}\right)\hat{j} + \left(\frac{\partial a_2}{\partial x} - \frac{\partial a_1}{\partial y}\right)\hat{k}$$

$$\operatorname{curl}\operatorname{curl} \mathbf{a} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial a_3}{\partial y} - \frac{\partial a_2}{\partial z} & \frac{\partial a_1}{\partial z} - \frac{\partial a_3}{\partial x} & \frac{\partial a_2}{\partial x} - \frac{\partial a_1}{\partial y} \end{vmatrix}$$

Expanding,

$$= \hat{i}\left[\frac{\partial}{\partial y}\left(\frac{\partial a_2}{\partial x} - \frac{\partial a_1}{\partial y}\right) - \frac{\partial}{\partial z}\left(\frac{\partial a_1}{\partial z} - \frac{\partial a_3}{\partial x}\right)\right] + \cdots$$

$$= \hat{i}\left[\frac{\partial^2 a_2}{\partial y\partial x} - \frac{\partial^2 a_1}{\partial y^2} - \frac{\partial^2 a_1}{\partial z^2} + \frac{\partial^2 a_3}{\partial z\partial x}\right] + \cdots$$

$$= \hat{i}\left[\frac{\partial}{\partial x}\left(\frac{\partial a_1}{\partial x} + \frac{\partial a_2}{\partial y} + \frac{\partial a_3}{\partial z}\right) - \left(\frac{\partial^2 a_1}{\partial x^2} + \frac{\partial^2 a_1}{\partial y^2} + \frac{\partial^2 a_1}{\partial z^2}\right)\right] + \cdots$$

$$= \nabla(\operatorname{div} \mathbf{a}) - \nabla^2\mathbf{a}$$

5. follows directly from (iv).

$$= \left(\hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}\right)(\operatorname{div}\mathbf{a}) - \nabla^2(a_1\hat{i} + a_2\hat{j} + a_3\hat{k})$$

$$= \operatorname{grad}(\operatorname{div}\mathbf{a}) - \nabla^2\mathbf{a}$$

**Example 2.2** If $F$ is a differentiable function of $x, y, z, t$ where $x, y, z$ are differentiable functions of $t$, prove that

$$\frac{dF}{dt} = \frac{\partial F}{\partial t} + (\mathbf{v}\cdot\nabla)F, \quad \mathbf{v} = \left(\frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt}\right)$$

**Solution.** We have

$$\begin{aligned}\frac{dF}{dt} &= \frac{\partial F}{\partial t} + \frac{dx}{dt}\frac{\partial F}{\partial x} + \frac{dy}{dt}\frac{\partial F}{\partial y} + \frac{dz}{dt}\frac{\partial F}{\partial z}\\ &= \frac{\partial F}{\partial t} + \left(\frac{dx}{dt}\frac{\partial}{\partial x} + \frac{dy}{dt}\frac{\partial}{\partial y} + \frac{dz}{dt}\frac{\partial}{\partial z}\right)F\\ &= \frac{\partial F}{\partial t} + (\mathbf{v}\cdot\nabla)F\end{aligned}$$

◀

**Example 2.3** Find the directional derivative of

$$\phi = x^2yz + 4xz^2$$

at $P(1, -2, -1)$ in the direction $2\hat{i} - \hat{j} - 2\hat{k}$.

**Solution.** We have

$$\begin{aligned}\nabla\phi &= \hat{i}\frac{\partial\phi}{\partial x} + \hat{j}\frac{\partial\phi}{\partial y} + \hat{k}\frac{\partial\phi}{\partial z}\\ &= (2xyz + 4z^2)\hat{i} + (x^2z)\hat{j} + (x^2y + 8xz)\hat{k}\end{aligned}$$

At $P(1, -2, -1)$:

$$\nabla\phi = 8\hat{i} - \hat{j} - 10\hat{k}$$

Unit vector in the direction $2\hat{i} - \hat{j} - 2\hat{k}$ is

$$\hat{a} = \frac{2\hat{i} - \hat{j} - 2\hat{k}}{3}$$

Directional derivative:

$$\nabla\phi\cdot\hat{a} = \frac{1}{3}(8\hat{i} - \hat{j} - 10\hat{k})\cdot(2\hat{i} - \hat{j} - 2\hat{k}) = \frac{37}{3}$$

◀

**Example 2.4** The directional derivative

$$\phi = axy^2 + byz + cx^3z^2$$

at the point $P(1,2,-1)$ has a maximum magnitude 64 in the direction parallel to $z$-axis. Determine the values of $a, b, c$.

**Solution.** Since grad $\phi$ is the maximum value of the directional derivative in magnitude and direction, we have

$$\nabla\phi = 64\hat{k} \quad \text{at } P(1,2,-1)$$

$$\Rightarrow \quad \frac{\partial\phi}{\partial x}\hat{\imath} + \frac{\partial\phi}{\partial y}\hat{\jmath} + \frac{\partial\phi}{\partial z}\hat{k} = 64\hat{k}$$

$$\Rightarrow \quad \frac{\partial\phi}{\partial x} = 0, \quad \frac{\partial\phi}{\partial y} = 0, \quad \frac{\partial\phi}{\partial z} = 64$$

$$\Rightarrow \quad ay^2 + 3cx^2z^2 = 0, \quad 2axy + bz = 0, \quad by + 2cx^3z = 64$$

At $P(1,2,-1)$:

$$4a + 3c = 0, \quad 4a - b = 0, \quad 2b - 2c = 64$$

Solving:

$$a = 6, \quad b = 24, \quad c = -8$$

◀

**Example 2.5** Find a unit normal vector to the surface

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

at the point $(1,1,1)$.

**Solution.** Let

$$\phi = \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1$$

A unit normal is

$$\hat{n} = \pm\frac{\nabla\phi}{|\nabla\phi|}$$

$$\nabla\phi = \frac{2x}{a^2}\hat{\imath} + \frac{2y}{b^2}\hat{\jmath} + \frac{2z}{c^2}\hat{k}$$

$$|\nabla\phi| = 2\sqrt{\frac{x^2}{a^4} + \frac{y^2}{b^4} + \frac{z^2}{c^4}}$$

Thus at $(1,1,1)$:

$$\hat{n} = \pm\frac{\frac{\hat{\imath}}{a^2} + \frac{\hat{\jmath}}{b^2} + \frac{\hat{k}}{c^2}}{\sqrt{\frac{1}{a^4} + \frac{1}{b^4} + \frac{1}{c^4}}}$$

◀

**Example 2.6** Find the vector components of a vector function $\mathbf{F}$ along the tangent and the outward normal to the surface $\phi(x, y, z) = 0$ at $P(x, y, z)$.

**Solution.** Since

$$\hat{n} = \frac{\nabla\phi}{|\nabla\phi|}$$

is the unit outward normal, the normal component is

$$\mathbf{F}_n = (\mathbf{F}\cdot\hat{n})\hat{n} = \frac{(\mathbf{F}\cdot\nabla\phi)\nabla\phi}{|\nabla\phi|^2}$$

◀

## 2.8 Exercises

1. Find the angle between the normals to the surfaces

$$x^2 + y^2 + z^2 = 9 \quad \text{and} \quad z = x^2 + y^2 - 3$$

   at the point $(2, -1, 2)$.

2. Prove that the lines drawn from a point on an ellipse to the foci make equal angles with the normal to the ellipse.

3. Find constants $a$ and $b$ such that the surface

$$ax^2 - byz = (a + 2)x$$

   is orthogonal to

$$4x^2 + y^2 + z^3 = 4$$

   at $(1, -1, 2)$.

4. Find div $\mathbf{a}$ and curl $\mathbf{a}$ if:

   (a) $\mathbf{a} = (-x\hat{\imath} + y\hat{\jmath})(x^2 + y^2)$

   (b) $\mathbf{a} = \nabla(x^3 + y^3 + z^3 - 3xyz)$

5. Show that for $\mathbf{a} = (-y\hat{\imath} + x\hat{\jmath})/(x^2 + y^2)$,

$$\text{div } \mathbf{a} = 0 \quad \text{and} \quad \text{curl } \mathbf{a} = 0.$$

6. Determine the constant $a$ so that the vector

$$(x + y)\hat{\imath} + (y - 2z)\hat{\jmath} + (x + az)\hat{k}$$

   is solenoidal.

7. Determine the constant $\lambda$ so that the vector

$$\mathbf{a} = xz^3\hat{\imath} - 2x^2yz\hat{\jmath} + \lambda yz^4\hat{k}$$

is irrotational at $(1, -1, 1)$.

8. If $\mathbf{r}$ is the position vector and $|\mathbf{r}| = r$, prove that:

   (a) $\nabla(r^n) = n\mathbf{r}r^{n-2}$

   (b) $\operatorname{div}(\mathbf{r} \times \nabla f(r)) = 0$

9. Prove that:

$$\nabla^2 f(r) = \frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{df}{dr}\right)$$

   Hence find $f(r)$ such that $\nabla^2 f(r) = 0$.

10. Find differential function $f(r)$ so that $f(r)\mathbf{r}$ is solenoidal.

11. If $\phi$ and $\psi$ are scalar functions, then prove that:

    (a) $\nabla \times (\phi\nabla\phi) = 0$

    (b) $\nabla \cdot (\phi\nabla\psi - \psi\nabla\phi) = \phi\nabla^2\psi - \psi\nabla^2\phi$

    (c) $\nabla \cdot (\phi\nabla\psi \times \nabla\phi) = 0$

    (d) $\nabla(\phi^n) = n\phi^{n-1}\nabla\phi$

Be careful about typos

# 3 Curvilinear Coordinates

Let the Cartesian coordinates $(x, y, z)$ of any point $P$ in space be expressed as functions of three continuously differentiable scalar variables $(u_1, u_2, u_3)$, so that

$$x = x(u_1, u_2, u_3), \quad y = y(u_1, u_2, u_3), \quad z = z(u_1, u_2, u_3). \tag{13}$$

Equivalently, if $\mathbf{r}$ denotes the position vector of $P$, then (13) can be written as

$$\mathbf{r} = \mathbf{r}(u_1, u_2, u_3). \tag{14}$$

Assuming that the relations (13) (or (14)) can be inverted to express $u_1, u_2, u_3$ in terms of $(x, y, z)$, we obtain

$$\begin{aligned} &u_1 = u_1(x, y, z), \quad u_2 = u_2(x, y, z), \quad u_3 = u_3(x, y, z), \\ \text{or} \quad &u_1 = u_1(\mathbf{r}), \quad u_2 = u_2(\mathbf{r}), \quad u_3 = u_3(\mathbf{r}). \end{aligned}$$

Thus, to any point $P(x, y, z)$, we can associate a unique triple $(u_1, u_2, u_3)$. Geometrically, the point $P$ is the point of intersection of the three surfaces

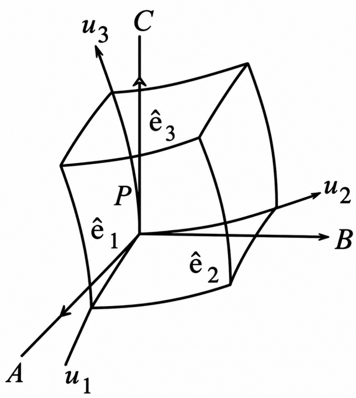
$$u_1 = c_1, \quad u_2 = c_2, \quad u_3 = c_3, \tag{15}$$

where $c_1, c_2, c_3$ are constants.

If the functions $u_1, u_2, u_3$ are such that the surfaces (15) neither coincide nor intersect along a common curve, then $(u_1, u_2, u_3)$ are called the *curvilinear coordinates* of the point $P$. The relations (13) or (14) define a *coordinate transformation*. The surfaces given by (15) are called *coordinate surfaces*. The curves formed by the intersection of each pair of these surfaces are called *coordinate curves* (or coordinate lines). The tangents to these curves at $P$ define the *coordinate axes* at that point. Since these coordinate axes depend on the point $P$, their directions vary from point to point. Consequently, the associated unit vectors are not constant but are functions of position.

## 3.1 Orthogonal Curvilinear Coordinates

A curvilinear coordinate system in which the coordinate surfaces intersect at right angles, i.e., the coordinate axes at every point are mutually perpendicular, is called an *orthogonal curvilinear coordinate system*. The corresponding coordinates are called *orthogonal curvilinear coordinates*.



Consider a right-handed orthogonal curvilinear coordinate system. Let $(u_1, u_2, u_3)$ be the coordinates of a point $P$. The pairs of coordinate surfaces at $P$,

$$u_2 = c_2,\ u_3 = c_3; \quad u_3 = c_3,\ u_1 = c_1; \quad u_1 = c_1,\ u_2 = c_2,$$

intersect to give the $u_1$-, $u_2$-, and $u_3$-curves, respectively.

Let the tangents to these curves at $P$ be $PA$, $PB$, and $PC$, which define the coordinate axes at $P$. Let $\hat{e}_1, \hat{e}_2, \hat{e}_3$ be the unit vectors along these axes. These form a right-handed orthonormal system analogous to $\hat{\imath}, \hat{\jmath}, \hat{k}$, with the distinction that $\hat{e}_1, \hat{e}_2, \hat{e}_3$ vary from point to point.

Thus,

$$\begin{gathered}
\hat{e}_1 \cdot \hat{e}_1 = \hat{e}_2 \cdot \hat{e}_2 = \hat{e}_3 \cdot \hat{e}_3 = 1, \\
\hat{e}_1 \cdot \hat{e}_2 = \hat{e}_2 \cdot \hat{e}_3 = \hat{e}_3 \cdot \hat{e}_1 = 0, \\
\hat{e}_1 \times \hat{e}_2 = \hat{e}_3, \quad \hat{e}_2 \times \hat{e}_3 = \hat{e}_1, \quad \hat{e}_3 \times \hat{e}_1 = \hat{e}_2.
\end{gathered} \tag{16}$$

Let $\mathbf{r}$ be the position vector of the point $P$. Then,

$$\mathbf{r} = \mathbf{r}(u_1, u_2, u_3),$$

and the differential displacement vector is

$$d\mathbf{r} = \frac{\partial \mathbf{r}}{\partial u_1}\, du_1 + \frac{\partial \mathbf{r}}{\partial u_2}\, du_2 + \frac{\partial \mathbf{r}}{\partial u_3}\, du_3. \tag{17}$$

Since $\frac{\partial \mathbf{r}}{\partial u_1}$, $\frac{\partial \mathbf{r}}{\partial u_2}$, and $\frac{\partial \mathbf{r}}{\partial u_3}$ are tangent to the $u_1$-, $u_2$-, and $u_3$-curves respectively, we can write

$$\frac{\partial \mathbf{r}}{\partial u_1} = h_1 \hat{e}_1,$$
$$\frac{\partial \mathbf{r}}{\partial u_2} = h_2 \hat{e}_2,$$
$$\frac{\partial \mathbf{r}}{\partial u_3} = h_3 \hat{e}_3,$$

where

$$h_1 = \left| \frac{\partial \mathbf{r}}{\partial u_1} \right|, \quad h_2 = \left| \frac{\partial \mathbf{r}}{\partial u_2} \right|, \quad h_3 = \left| \frac{\partial \mathbf{r}}{\partial u_3} \right|$$

are called the *scale factors*.

Hence, equation (17) becomes

$$d\mathbf{r} = h_1 \hat{e}_1\, du_1 + h_2 \hat{e}_2\, du_2 + h_3 \hat{e}_3\, du_3. \tag{18}$$

Any vector $\mathbf{a}$ in this coordinate system can be expressed as

$$\mathbf{a} = a_1 \hat{e}_1 + a_2 \hat{e}_2 + a_3 \hat{e}_3,$$

where $a_1, a_2, a_3$ are its components.

## 3.2 Differential Arc Length and Volume Element

Using (18) and the orthogonality relations (16), the square of the differential arc length is

$$ds^2 = d\mathbf{r} \cdot d\mathbf{r} = h_1^2\, du_1^2 + h_2^2\, du_2^2 + h_3^2\, du_3^2.$$

Along the $u_1$-curve (with $u_2, u_3$ constant),

$$d\mathbf{r} = h_1 \hat{e}_1\, du_1,$$

so the differential arc length is $ds_1 = h_1\, du_1$. Similarly,

$$ds_1 = h_1\, du_1, \quad ds_2 = h_2\, du_2, \quad ds_3 = h_3\, du_3.$$

For an elementary parallelepiped at $P$, the areas of the faces are

$$dS_1 = h_2 h_3\, du_2\, du_3, \quad dS_2 = h_3 h_1\, du_3\, du_1, \quad dS_3 = h_1 h_2\, du_1\, du_2,$$

and the volume element is

$$dV = h_1 h_2 h_3\, du_1\, du_2\, du_3. \tag{19}$$

## 3.3 Gradient

Let $\phi(u_1, u_2, u_3)$ be a scalar function. Then

$$\nabla\phi = \lambda_1 \hat{e}_1 + \lambda_2 \hat{e}_2 + \lambda_3 \hat{e}_3$$

where $\lambda_1, \lambda_2, \lambda_3$ are to be determined.

Since

$$d\mathbf{r} = \hat{e}_1 h_1 du_1 + \hat{e}_2 h_2 du_2 + \hat{e}_3 h_3 du_3,$$

we have

$$d\phi = \nabla\phi \cdot d\mathbf{r} = \lambda_1 h_1 du_1 + \lambda_2 h_2 du_2 + \lambda_3 h_3 du_3 \tag{20}$$

Also,

$$d\phi = \frac{\partial\phi}{\partial u_1} du_1 + \frac{\partial\phi}{\partial u_2} du_2 + \frac{\partial\phi}{\partial u_3} du_3 \tag{21}$$

Comparing (20) and (21), we get

$$\lambda_1 h_1 = \frac{\partial\phi}{\partial u_1}, \quad \lambda_2 h_2 = \frac{\partial\phi}{\partial u_2}, \quad \lambda_3 h_3 = \frac{\partial\phi}{\partial u_3}$$

Thus,

$$\nabla\phi = \frac{\hat{e}_1}{h_1}\frac{\partial\phi}{\partial u_1} + \frac{\hat{e}_2}{h_2}\frac{\partial\phi}{\partial u_2} + \frac{\hat{e}_3}{h_3}\frac{\partial\phi}{\partial u_3} \tag{22}$$

### 3.3.1 Operator $\nabla$

Equation (22) can be written as

$$\nabla = \frac{\hat{e}_1}{h_1}\frac{\partial}{\partial u_1} + \frac{\hat{e}_2}{h_2}\frac{\partial}{\partial u_2} + \frac{\hat{e}_3}{h_3}\frac{\partial}{\partial u_3} \tag{23}$$

### 3.3.2 Unit vectors

We already have

$$\hat{e}_1 = \frac{1}{h_1}\frac{\partial \mathbf{r}}{\partial u_1}, \quad \hat{e}_2 = \frac{1}{h_2}\frac{\partial \mathbf{r}}{\partial u_2}, \quad \hat{e}_3 = \frac{1}{h_3}\frac{\partial \mathbf{r}}{\partial u_3}$$

From (22), we can write

$$\nabla u_1 = \frac{\hat{e}_1}{h_1}, \quad \nabla u_2 = \frac{\hat{e}_2}{h_2}, \quad \nabla u_3 = \frac{\hat{e}_3}{h_3}$$

Hence,

$$\hat{e}_1 = h_1 \nabla u_1, \quad \hat{e}_2 = h_2 \nabla u_2, \quad \hat{e}_3 = h_3 \nabla u_3 \tag{24}$$

Using vector identities,

$$\begin{aligned} \hat{e}_1 &= \hat{e}_2 \times \hat{e}_3 = h_2 h_3 \nabla u_2 \times \nabla u_3 \\ \hat{e}_2 &= \hat{e}_3 \times \hat{e}_1 = h_3 h_1 \nabla u_3 \times \nabla u_1 \\ \hat{e}_3 &= \hat{e}_1 \times \hat{e}_2 = h_1 h_2 \nabla u_1 \times \nabla u_2 \end{aligned} \tag{25}$$

## 3.4 Divergence

Let $\mathbf{a}(u_1, u_2, u_3)$ be a vector field such that

$$\mathbf{a} = a_1 \hat{e}_1 + a_2 \hat{e}_2 + a_3 \hat{e}_3$$

Then, using (25), we can write

$$\mathbf{a} = a_1 h_2 h_3 \nabla u_2 \times \nabla u_3 + a_2 h_3 h_1 \nabla u_3 \times \nabla u_1 + a_3 h_1 h_2 \nabla u_1 \times \nabla u_2$$

Therefore, using the vector identities, we get

$$\begin{aligned} \nabla \cdot \mathbf{a} &= \nabla \cdot (a_1 h_2 h_3 \nabla u_2 \times \nabla u_3) + \cdots \\ &= a_1 h_2 h_3 \nabla \cdot (\nabla u_2 \times \nabla u_3) + \nabla(a_1 h_2 h_3) \cdot (\nabla u_2 \times \nabla u_3) + \cdots \end{aligned}$$

Using the identity

$$\nabla \cdot (\nabla u_2 \times \nabla u_3) = \nabla u_3 \cdot (\nabla \times \nabla u_2) - \nabla u_2 \cdot (\nabla \times \nabla u_3) = 0,$$

we obtain

$$\nabla \cdot \mathbf{a} = (\nabla u_2 \times \nabla u_3) \cdot \nabla(a_1 h_2 h_3) + \cdots$$

Using relations from (24), we get

$$\nabla u_2 \times \nabla u_3 = \frac{\hat{e}_1}{h_2 h_3}$$

Thus,

$$\nabla \cdot \mathbf{a} = \frac{\hat{e}_1}{h_2 h_3} \cdot \nabla(a_1 h_2 h_3) + \cdots$$

Expanding similarly for all terms, we obtain

$$\nabla \cdot \mathbf{a} = \frac{1}{h_1 h_2 h_3}\left[\frac{\partial}{\partial u_1}(a_1 h_2 h_3) + \frac{\partial}{\partial u_2}(a_2 h_3 h_1) + \frac{\partial}{\partial u_3}(a_3 h_1 h_2)\right] \tag{26}$$

### 3.5 Curl

Let $\mathbf{a} = a(u_1, u_2, u_3)$ be a vector function. Using (24), we write

$$\mathbf{a} = a_1\hat{e}_1 + a_2\hat{e}_2 + a_3\hat{e}_3 = a_1h_1\nabla u_1 + a_2h_2\nabla u_2 + a_3h_3\nabla u_3$$

Using vector identities,

$$\begin{aligned}
\nabla \times \mathbf{a} &= \nabla \times (a_1h_1\nabla u_1) + \cdots \\
&= \nabla(a_1h_1) \times \nabla u_1 + \cdots \\
\nabla \times \mathbf{a} &= \left[\frac{\hat{e}_1}{h_1}\frac{\partial}{\partial u_1} + \frac{\hat{e}_2}{h_2}\frac{\partial}{\partial u_2} + \frac{\hat{e}_3}{h_3}\frac{\partial}{\partial u_3}\right] \times (a_1h_1\nabla u_1 + \cdots)
\end{aligned}$$

After simplification, we obtain

$$\nabla \times \mathbf{a} = \frac{1}{h_1h_2h_3}\begin{vmatrix} h_1\hat{e}_1 & h_2\hat{e}_2 & h_3\hat{e}_3 \\ \dfrac{\partial}{\partial u_1} & \dfrac{\partial}{\partial u_2} & \dfrac{\partial}{\partial u_3} \\ a_1h_1 & a_2h_2 & a_3h_3 \end{vmatrix} \tag{27}$$

### 3.6 Laplacian $\nabla^2$

Let $\phi = \phi(u_1, u_2, u_3)$ be a scalar function. Then

$$\nabla^2\phi = \nabla \cdot (\nabla\phi)$$

Using the expression for gradient,

$$\nabla\phi = \frac{\hat{e}_1}{h_1}\frac{\partial\phi}{\partial u_1} + \frac{\hat{e}_2}{h_2}\frac{\partial\phi}{\partial u_2} + \frac{\hat{e}_3}{h_3}\frac{\partial\phi}{\partial u_3}$$

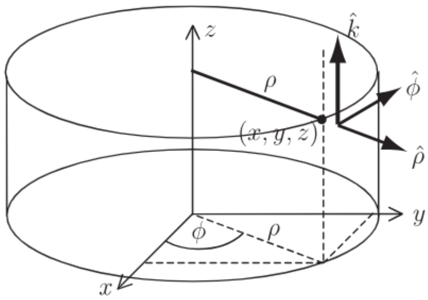
we obtain

$$\nabla^2\phi = \frac{1}{h_1h_2h_3}\left[\frac{\partial}{\partial u_1}\left(\frac{h_2h_3}{h_1}\frac{\partial\phi}{\partial u_1}\right) + \frac{\partial}{\partial u_2}\left(\frac{h_3h_1}{h_2}\frac{\partial\phi}{\partial u_2}\right) + \frac{\partial}{\partial u_3}\left(\frac{h_1h_2}{h_3}\frac{\partial\phi}{\partial u_3}\right)\right] \tag{28}$$

### 3.7 Cylindrical Polar Coordinates

The transformation equations connecting Cartesian coordinates $(x, y, z)$ and cylindrical polar coordinates $(\rho, \phi, z)$ are

$$x = \rho\cos\phi, \quad y = \rho\sin\phi, \quad z = z. \tag{29}$$




with

$$\rho \geq 0, \quad 0 \leq \phi \leq 2\pi, \quad -\infty < z < \infty.$$

The position vector $\mathbf{r}$ of a point $P$ is

$$\begin{aligned}\mathbf{r} &= x\hat{\imath} + y\hat{\jmath} + z\hat{k}\\ &= (\rho\cos\phi)\hat{\imath} + (\rho\sin\phi)\hat{\jmath} + z\hat{k}.\end{aligned}$$

The differential displacement vector is

$$\begin{aligned}d\mathbf{r} &= \frac{\partial \mathbf{r}}{\partial \rho}d\rho + \frac{\partial \mathbf{r}}{\partial \phi}d\phi + \frac{\partial \mathbf{r}}{\partial z}dz\\ &= (\hat{\imath}\cos\phi + \hat{\jmath}\sin\phi)\,d\rho + (-\hat{\imath}\rho\sin\phi + \hat{\jmath}\rho\cos\phi)\,d\phi + \hat{k}\,dz\\ &= (\cos\phi\,d\rho - \rho\sin\phi\,d\phi)\hat{\imath} + (\sin\phi\,d\rho + \rho\cos\phi\,d\phi)\hat{\jmath} + dz\,\hat{k}.\end{aligned}$$

Hence, the square of the differential arc length is

$$ds^2 = d\mathbf{r}\cdot d\mathbf{r} = d\rho^2 + \rho^2 d\phi^2 + dz^2. \tag{30}$$

Thus, in cylindrical coordinates,

$$u_1 = \rho, \quad u_2 = \phi, \quad u_3 = z,$$

with corresponding unit vectors

$$\hat{e}_1 = \hat{\rho}, \quad \hat{e}_2 = \hat{\phi}, \quad \hat{e}_3 = \hat{k},$$

and scale factors

$$h_1 = 1, \quad h_2 = \rho, \quad h_3 = 1. \tag{31}$$

Let $f(\rho,\phi,z)$ be a scalar function and

$$\mathbf{a} = a_\rho\hat{\rho} + a_\phi\hat{\phi} + a_z\hat{k}$$

be a vector field. Then:

**Gradient**

$$\nabla f = \hat{\rho}\frac{\partial f}{\partial \rho} + \hat{\phi}\frac{1}{\rho}\frac{\partial f}{\partial \phi} + \hat{k}\frac{\partial f}{\partial z}. \tag{32}$$

**Divergence**

$$\nabla\cdot\mathbf{a} = \frac{1}{\rho}\frac{\partial}{\partial \rho}(\rho a_\rho) + \frac{1}{\rho}\frac{\partial a_\phi}{\partial \phi} + \frac{\partial a_z}{\partial z}. \tag{33}$$

**Curl**

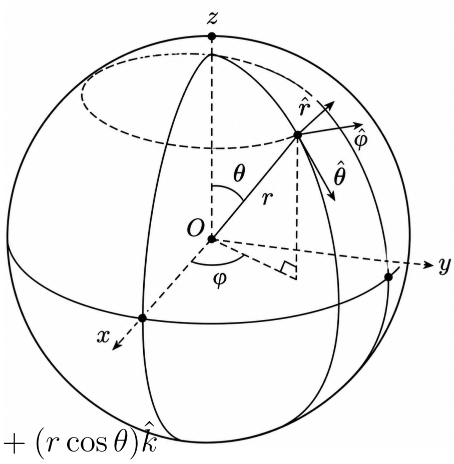
$$\nabla\times\mathbf{a} = \frac{1}{\rho}\begin{vmatrix}\hat{\rho} & \rho\hat{\phi} & \hat{k}\\ \frac{\partial}{\partial \rho} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z}\\ a_\rho & \rho a_\phi & a_z\end{vmatrix}. \tag{34}$$

**Laplacian**

$$\nabla^2 f = \frac{1}{\rho}\frac{\partial}{\partial \rho}\left(\rho\frac{\partial f}{\partial \rho}\right) + \frac{1}{\rho^2}\frac{\partial^2 f}{\partial \phi^2} + \frac{\partial^2 f}{\partial z^2}. \tag{35}$$

## 3.8 Spherical Polar Coordinates

The transformation equations connecting the Cartesian coordinates $(x, y, z)$ and the spherical polar coordinates $(r, \theta, \phi)$ are given by




$$x = r\sin\theta\cos\phi, \quad y = r\sin\theta\sin\phi, \quad z = r\cos\theta$$

$$r \geq 0, \quad 0 \leq \theta \leq \pi, \quad 0 \leq \phi \leq 2\pi$$

Hence, if $\mathbf{r}$ is the position vector of the point $P$, then

$$\mathbf{r} = (r\sin\theta\cos\phi)\hat{\imath} + (r\sin\theta\sin\phi)\hat{\jmath} + (r\cos\theta)\hat{k}$$

$$d\mathbf{r} = \frac{\partial \mathbf{r}}{\partial r}dr + \frac{\partial \mathbf{r}}{\partial \theta}d\theta + \frac{\partial \mathbf{r}}{\partial \phi}d\phi$$

$$= (\hat{\imath}\sin\theta\cos\phi + \hat{\jmath}\sin\theta\sin\phi + \hat{k}\cos\theta)dr$$
$$+(r\hat{\imath}\cos\theta\cos\phi + r\hat{\jmath}\cos\theta\sin\phi - r\hat{k}\sin\theta)d\theta$$
$$+(-r\hat{\imath}\sin\theta\sin\phi + r\hat{\jmath}\sin\theta\cos\phi)d\phi$$

$$ds^2 = d\mathbf{r}\cdot d\mathbf{r} = (dr)^2 + r^2(d\theta)^2 + r^2\sin^2\theta(d\phi)^2$$

Hence, for spherical coordinates,

$$u_1 = r, \quad u_2 = \theta, \quad u_3 = \phi; \quad \hat{e}_1 = \hat{r},\ \hat{e}_2 = \hat{\theta},\ \hat{e}_3 = \hat{\phi}$$

$$h_1 = 1, \quad h_2 = r, \quad h_3 = r\sin\theta$$

Thus, for scalar function $f(r, \theta, \phi)$ and vector function

$$\mathbf{a} = a_r\hat{r} + a_\theta\hat{\theta} + a_\phi\hat{\phi}$$

$$\nabla f = \hat{r}\frac{\partial f}{\partial r} + \hat{\theta}\frac{1}{r}\frac{\partial f}{\partial \theta} + \hat{\phi}\frac{1}{r\sin\theta}\frac{\partial f}{\partial \phi}$$

$$\nabla\cdot\mathbf{a} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 a_r) + \frac{1}{r\sin\theta}\frac{\partial}{\partial \theta}(\sin\theta\, a_\theta) + \frac{1}{r\sin\theta}\frac{\partial a_\phi}{\partial \phi}$$

$$\nabla\times\mathbf{a} = \frac{1}{r\sin\theta}\begin{vmatrix} \hat{r} & r\hat{\theta} & r\sin\theta\hat{\phi} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ a_r & ra_\theta & r\sin\theta a_\phi \end{vmatrix} \tag{36}$$

$$\nabla^2 f = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial f}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta\frac{\partial f}{\partial \theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2 f}{\partial \phi^2}$$

## 3.9 Alternative Method for Finding the Scale Factors and the Unit Vectors

If $(u_1, u_2, u_3)$ be the position vector of a point $P$, then ,

$$\frac{\partial \mathbf{r}}{\partial u_1} = h_1 \hat{e}_1, \quad \frac{\partial \mathbf{r}}{\partial u_2} = h_2 \hat{e}_2, \quad \frac{\partial \mathbf{r}}{\partial u_3} = h_3 \hat{e}_3$$

and therefore, we get

$$h_1 = \left|\frac{\partial \mathbf{r}}{\partial u_1}\right|, \quad h_2 = \left|\frac{\partial \mathbf{r}}{\partial u_2}\right|, \quad h_3 = \left|\frac{\partial \mathbf{r}}{\partial u_3}\right| \tag{37}$$

$$\hat{e}_1 = \frac{\partial \mathbf{r}/\partial u_1}{h_1}, \quad \hat{e}_2 = \frac{\partial \mathbf{r}/\partial u_2}{h_2}, \quad \hat{e}_3 = \frac{\partial \mathbf{r}/\partial u_3}{h_3}$$

### 3.9.1 Cylindrical Coordinates

Since

$$\begin{aligned}
\mathbf{r} &= (\rho\cos\phi)\hat{\imath} + (\rho\sin\phi)\hat{\jmath} + z\hat{k} \\
\frac{\partial \mathbf{r}}{\partial \rho} &= \hat{\imath}\cos\phi + \hat{\jmath}\sin\phi, \quad \frac{\partial \mathbf{r}}{\partial \phi} = -\hat{\imath}\rho\sin\phi + \hat{\jmath}\rho\cos\phi, \quad \frac{\partial \mathbf{r}}{\partial z} = \hat{k}
\end{aligned}$$

Hence, we get by (37)

$$h_1 = \left|\frac{\partial \mathbf{r}}{\partial \rho}\right| = 1, \quad h_2 = \left|\frac{\partial \mathbf{r}}{\partial \phi}\right| = \rho, \quad h_3 = \left|\frac{\partial \mathbf{r}}{\partial z}\right| = 1$$

$$\hat{e}_1 = \hat{\rho} = \hat{\imath}\cos\phi + \hat{\jmath}\sin\phi, \quad \hat{e}_2 = \hat{\phi} = -\hat{\imath}\sin\phi + \hat{\jmath}\cos\phi, \quad \hat{e}_3 = \hat{k}$$

### 3.9.2 Spherical Coordinates

Since

$$\begin{aligned}
\mathbf{r} &= (r\sin\theta\cos\phi)\hat{\imath} + (r\sin\theta\sin\phi)\hat{\jmath} + (r\cos\theta)\hat{k} \\
\frac{\partial \mathbf{r}}{\partial r} &= \sin\theta(\hat{\imath}\cos\phi + \hat{\jmath}\sin\phi) + \hat{k}\cos\theta \\
\frac{\partial \mathbf{r}}{\partial \theta} &= r\cos\theta(\hat{\imath}\cos\phi + \hat{\jmath}\sin\phi) - r\hat{k}\sin\theta \\
\frac{\partial \mathbf{r}}{\partial \phi} &= r\sin\theta(-\hat{\imath}\sin\phi + \hat{\jmath}\cos\phi)
\end{aligned}$$

Hence, we obtain by (37)

$$\begin{aligned}
h_1 &= \left|\frac{\partial \mathbf{r}}{\partial r}\right| = 1, \quad \hat{e}_1 = \hat{r} = \sin\theta(\hat{\imath}\cos\phi + \hat{\jmath}\sin\phi) + \hat{k}\cos\theta \\
h_2 &= \left|\frac{\partial \mathbf{r}}{\partial \theta}\right| = r, \quad \hat{e}_2 = \hat{\theta} = \cos\theta(\hat{\imath}\cos\phi + \hat{\jmath}\sin\phi) - \hat{k}\sin\theta \\
h_3 &= \left|\frac{\partial \mathbf{r}}{\partial \phi}\right| = r\sin\theta, \quad \hat{e}_3 = \hat{\phi} = -\hat{\imath}\sin\phi + \hat{\jmath}\cos\phi
\end{aligned}$$

**Example 3.1** Find the scale factors and the unit vectors of the parabolic cylindrical coordinate system:

$$x = \frac{1}{2}(u^2 - v^2), \quad y = uv, \quad z = z$$

$$-\infty < u < \infty, \quad v \geq 0, \quad -\infty < z < \infty$$

and obtain the expression for the gradient of the scalar function $f(u, v, z)$ in this system.

**Solution.** We have

$$\mathbf{r} = \frac{1}{2}(u^2 - v^2)\hat{\imath} + uv\,\hat{\jmath} + z\hat{k}$$

so that

$$\frac{\partial \mathbf{r}}{\partial u} = u\hat{\imath} + v\hat{\jmath}, \quad \frac{\partial \mathbf{r}}{\partial v} = -v\hat{\imath} + u\hat{\jmath}, \quad \frac{\partial \mathbf{r}}{\partial z} = \hat{k}$$

$$h_1 = \left|\frac{\partial \mathbf{r}}{\partial u}\right| = \sqrt{u^2 + v^2}, \quad \hat{e}_1 = \hat{u} = \frac{u\hat{\imath} + v\hat{\jmath}}{\sqrt{u^2 + v^2}}$$

$$h_2 = \left|\frac{\partial \mathbf{r}}{\partial v}\right| = \sqrt{u^2 + v^2}, \quad \hat{e}_2 = \hat{v} = \frac{-v\hat{\imath} + u\hat{\jmath}}{\sqrt{u^2 + v^2}}$$

$$h_3 = \left|\frac{\partial \mathbf{r}}{\partial z}\right| = 1, \quad \hat{e}_3 = \hat{k}$$

Then, by (22),

$$\nabla f = \frac{\hat{u}}{\sqrt{u^2 + v^2}}\frac{\partial f}{\partial u} + \frac{\hat{v}}{\sqrt{u^2 + v^2}}\frac{\partial f}{\partial v} + \hat{k}\frac{\partial f}{\partial z}$$

◀

**Example 3.2** Prove that

1. In cylindrical coordinates $(\rho, \phi, z)$, $\nabla \cdot (\nabla \log \rho) = 0$.
2. In spherical coordinates $(r, \theta, \phi)$, $\nabla \times \hat{\theta} = \dfrac{\cos\theta}{r^2}\hat{r}$.

**Solution.** **(1.)** In cylindrical coordinates,

$$\nabla \log \rho = \hat{\rho}\frac{\partial}{\partial \rho}(\log \rho) + \hat{\phi}\frac{1}{\rho}\frac{\partial}{\partial \phi}(\log \rho) + \hat{k}\frac{\partial}{\partial z}(\log \rho) = \frac{1}{\rho}\hat{\rho}$$

Then, we get

$$\nabla \cdot (\nabla \log \rho) = \frac{1}{\rho}\frac{\partial}{\partial \rho}\left(\rho \cdot \frac{1}{\rho}\right) + \frac{1}{\rho}\frac{\partial}{\partial \phi}(0) + \frac{\partial}{\partial z}(0) = \frac{1}{\rho}\frac{\partial}{\partial \rho}(1) = 0$$

**2.** By (36), in spherical coordinates,

$$\nabla \times \hat{\theta} = \frac{1}{r^2 \sin\theta} \begin{vmatrix} \hat{r} & r\hat{\theta} & r\sin\theta\hat{\phi} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ 0 & r & 0 \end{vmatrix}$$

$$= \frac{1}{r^2 \sin\theta} \begin{vmatrix} \hat{r} & r\hat{\theta} & r\sin\theta\hat{\phi} \\ 0 & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ 0 & r & 0 \end{vmatrix} = \frac{1}{r}\hat{\phi}$$

More precisely,

$$\nabla \times \hat{\theta} = \frac{\cos\theta}{r^2}\hat{r}$$

◀

**Example 3.3** If $u_1, u_2, u_3$ are orthogonal curvilinear coordinates, then show that

$$\nabla u_1,\ \nabla u_2,\ \nabla u_3$$

are reciprocal systems of vectors.

**Solution.** We know that

$$\frac{\partial \mathbf{r}}{\partial u_1} = h_1\hat{e}_1, \quad \frac{\partial \mathbf{r}}{\partial u_2} = h_2\hat{e}_2, \quad \frac{\partial \mathbf{r}}{\partial u_3} = h_3\hat{e}_3$$

$$\nabla u_1 = \frac{\hat{e}_1}{h_1}, \quad \nabla u_2 = \frac{\hat{e}_2}{h_2}, \quad \nabla u_3 = \frac{\hat{e}_3}{h_3}$$

Consequently,

$$\frac{\partial \mathbf{r}}{\partial u_1} \cdot \nabla u_1 = 1, \quad \frac{\partial \mathbf{r}}{\partial u_1} \cdot \nabla u_2 = 0, \quad \frac{\partial \mathbf{r}}{\partial u_1} \cdot \nabla u_3 = 0$$

$$\frac{\partial \mathbf{r}}{\partial u_2} \cdot \nabla u_1 = 0, \quad \frac{\partial \mathbf{r}}{\partial u_2} \cdot \nabla u_2 = 1, \quad \frac{\partial \mathbf{r}}{\partial u_2} \cdot \nabla u_3 = 0$$

$$\frac{\partial \mathbf{r}}{\partial u_3} \cdot \nabla u_1 = 0, \quad \frac{\partial \mathbf{r}}{\partial u_3} \cdot \nabla u_2 = 0, \quad \frac{\partial \mathbf{r}}{\partial u_3} \cdot \nabla u_3 = 1$$

Hence, $\nabla u_1, \nabla u_2, \nabla u_3$ form a reciprocal system.

$$\frac{\partial r}{\partial u_3} \cdot \nabla u_1 = 0, \quad \frac{\partial r}{\partial u_3} \cdot \nabla u_2 = 0, \quad \frac{\partial r}{\partial u_3} \cdot \nabla u_3 = 1$$

Hence the given systems are the reciprocal systems. ◀

## 3.10 Exercises

1. Find the scale factors and the unit vectors for the following coordinate systems:

   (a) *Paraboloidal coordinates* $(u, v, \phi)$:

   $$x = uv\cos\phi, \quad y = uv\sin\phi, \quad z = \frac{1}{2}(u^2 - v^2)$$
   $$u \geq 0, \; v \geq 0, \; 0 \leq \phi \leq 2\pi$$

   (b) *Elliptic cylindrical coordinates* $(u, v, z)$:

   $$x = a\cosh u\cos v, \quad y = a\sinh u\sin v, \quad z = z$$
   $$u \geq 0, \; 0 \leq v \leq 2\pi, \; -\infty < z < \infty$$

2. Obtain expressions for the operator $\nabla$, div $\mathbf{a}$, curl $\mathbf{a}$ and $\nabla^2 f$ for the scalar function $f(u, v, z)$ and the vector function $\mathbf{a}(u, v, z)$ in the parabolic cylindrical coordinates

   $$x = \frac{1}{2}(u^2 - v^2), \quad y = uv, \quad z = z$$

3. Find the expressions for the grad $f$, $\nabla$, div $\mathbf{a}$, curl $\mathbf{a}$ and $\nabla^2 f$ for the scalar function $f$ and the vector function $\mathbf{a}$, in the coordinate systems defined in the question 1.

4. Show that the cylindrical coordinate system and the spherical coordinate system are orthogonal.

5. Find the volume elements in (i) cylindrical, (ii) spherical and (iii) parabolic cylindrical coordinates.

6. If $u_1, u_2, u_3$ are orthogonal curvilinear coordinates, then prove that

   $$\begin{vmatrix} \dfrac{\partial r}{\partial u_1} & \dfrac{\partial r}{\partial u_2} & \dfrac{\partial r}{\partial u_3} \end{vmatrix} \cdot \begin{vmatrix} \nabla u_1 & \nabla u_2 & \nabla u_3 \end{vmatrix} = 1$$

7. Express the unit vectors $\hat{\imath}, \hat{\jmath}, \hat{k}$ of the cartesian coordinate system in terms of the unit vectors $\hat{\rho}, \hat{\phi}, \hat{k}$ of the cylindrical coordinate system and also in terms of the unit vectors $\hat{r}, \hat{\theta}, \hat{\phi}$ of the spherical coordinate system.

8. Show that, if dots denote differentiation with respect to time $t$, then

   (a) in cylindrical polar coordinates,

   $$\dot{\hat{\rho}} = \dot{\phi}\,\hat{\phi}, \quad \dot{\hat{\phi}} = -\dot{\phi}\,\hat{\rho}$$

   (b) in spherical polar coordinates,

   $$\dot{\hat{r}} = \dot{\theta}\hat{\theta} + \sin\theta\dot{\phi}\hat{\phi}, \quad \dot{\hat{\theta}} = -\dot{\theta}\hat{r} + \cos\theta\dot{\phi}\hat{\phi}, \quad \dot{\hat{\phi}} = -\sin\theta\dot{\phi}\hat{r} - \cos\theta\dot{\phi}\hat{\theta}$$

9. Obtain the expressions for velocity and acceleration of a particle in (i) cylindrical (ii) spherical coordinates.

Be careful about typos

# 4 Vector Integration

We are familiar with the concept of integration of scalar functions as studied in real analysis. In this chapter, we introduce the concept of integration of vector functions and define line, surface, and volume integrals. We also study their transformations through the theorems of Gauss, Green, and Stokes. The definitions of these integrals are analogous to those in ordinary integral calculus.

Let $\mathbf{F}(t)$ be a continuously differentiable vector function of a scalar variable $t$ in a specified region, and let $\mathbf{f}(t)$ be its derivative, i.e.,

$$\frac{d\mathbf{F}(t)}{dt} = \mathbf{f}(t).$$

Then, the indefinite integral of $\mathbf{f}(t)$ is defined as

$$\int \mathbf{f}(t)\,dt = \mathbf{F}(t) + \mathbf{c}, \tag{38}$$

where $\mathbf{c}$ is an arbitrary constant vector independent of $t$.

With this definition (38), the definite integral of $\mathbf{f}(t)$ between the limits $t = a$ and $t = b$ is given by

$$\int_a^b \mathbf{f}(t)\,dt = \mathbf{F}(b) - \mathbf{F}(a).$$

Alternatively, the definite integral may be defined as the limit of a sum, analogous to ordinary calculus:

$$\int_a^b \mathbf{f}(t)\,dt = \lim_{n\to\infty} \sum_{i=1}^{n} \mathbf{f}(\xi_i)\,\Delta t_i, \tag{39}$$

where $\xi_i \in [t_{i-1}, t_i]$ and $\Delta t_i = t_i - t_{i-1}$ for a partition

$$a = t_0 < t_1 < \cdots < t_n = b.$$

## 4.1 Line Integral

Consider an oriented curve $C$ in space. Assume that $C$ is a simple curve, i.e., it does not intersect or touch itself. Let $P_0$ be the initial point and $P$ the terminal point of $C$ with respect to the chosen orientation. Let the parametric representation of $C$, with arc length $s$ as parameter, be

$$\mathbf{r} = \mathbf{r}(s) = x(s)\hat{\imath} + y(s)\hat{\jmath} + z(s)\hat{k}. \tag{40}$$

Assume that $\mathbf{r}(s)$ is continuous and possesses a continuous first derivative which is non-zero for all $s$. Then $C$ is a smooth curve, i.e., it has a unique tangent at each point.

Let $\mathbf{F}$ be a continuous vector point function defined on $C$. Divide $C$ into $n$ segments by points $P_0, P_1, \ldots, P_n = P$ with position vectors $\mathbf{r}_0, \mathbf{r}_1, \ldots, \mathbf{r}_n$, and define

$$\Delta\mathbf{r}_i = \mathbf{r}_i - \mathbf{r}_{i-1}, \quad i = 1, 2, \ldots, n.$$

Let $Q_i$ be any point on the segment $P_{i-1}P_i$. Then form the sum

$$I_n = \sum_{i=1}^{n} \mathbf{F}(Q_i) \cdot \Delta\mathbf{r}_i.$$

If the limit exists as the maximum length of the segments tends to zero, then

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \lim_{n\to\infty} \sum_{i=1}^{n} \mathbf{F}(Q_i) \cdot \Delta\mathbf{r}_i, \tag{41}$$

which is called the *line integral* of $\mathbf{F}$ along $C$ from $P_0$ to $P$.

If $\mathbf{F} = F_1\hat{\imath} + F_2\hat{\jmath} + F_3\hat{k}$, then

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C (F_1\, dx + F_2\, dy + F_3\, dz). \tag{42}$$

Since the unit tangent vector to $C$ is $\hat{\mathbf{t}} = \dfrac{d\mathbf{r}}{ds}$, we have

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C \mathbf{F} \cdot \frac{d\mathbf{r}}{ds}\, ds = \int_C \mathbf{F} \cdot \hat{\mathbf{t}}\, ds. \tag{43}$$

Thus, the line integral represents the integral of the tangential component of $\mathbf{F}$ along the curve and is therefore called the *tangential line integral*.

If the curve $C$ is parametrized by $t$, i.e., $\mathbf{r} = \mathbf{r}(t)$, then

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_{t_0}^{t_1} \mathbf{F}(\mathbf{r}(t)) \cdot \frac{d\mathbf{r}}{dt}\, dt. \tag{44}$$

In general, the value of a line integral depends on the path between the endpoints.

### 4.1.1 Physical Interpretation

If a variable force $\mathbf{F}$ acts on a particle moving along a curve $C$, then the work done is given by

$$W = \int_C \mathbf{F} \cdot d\mathbf{r}. \tag{45}$$

### 4.1.2 Line Integrals Independent of Path

Let $\mathbf{F}$ be a continuous vector field defined in a region $D$. The line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ is said to be *independent of path* in $D$ if its value depends only on the endpoints and not on the path joining them.

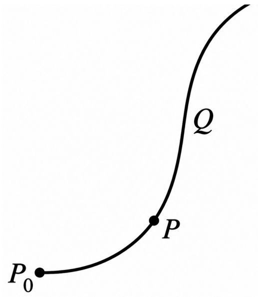
**Theorem 4.1.1.** *The line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of path in a region $D$ if and only if there exists a scalar function $\phi$ such that*

$$\mathbf{F} = \nabla\phi \tag{46}$$

*in $D$.*

*Proof.* Assume the line integral is independent of path. Fix a point $P_0$ and define

$$\phi(P) = \int_{P_0}^{P} \mathbf{F} \cdot d\mathbf{r}.$$




Assume the line integral is independent of path. Fix a point $P_0$ and define

$$\phi(P) = \int_{P_0}^{P} \mathbf{F} \cdot d\mathbf{r}.$$

For a nearby point $Q$,

$$\phi(Q) - \phi(P) = \int_{P}^{Q} \mathbf{F} \cdot d\mathbf{r}.$$

Thus,

$$d\phi = \mathbf{F} \cdot d\mathbf{r} \quad \Rightarrow \quad \nabla\phi \cdot d\mathbf{r} = \mathbf{F} \cdot d\mathbf{r}.$$

Since $d\mathbf{r}$ is arbitrary, it follows that

$$\mathbf{F} = \nabla\phi.$$

Conversely, if $\mathbf{F} = \nabla\phi$, then

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C \nabla\phi \cdot d\mathbf{r} = \int_{P_0}^{P} d\phi = \phi(P) - \phi(P_0),$$

which depends only on the endpoints and is therefore independent of the path. □

#### 4.1.3 Circulation and Irrotational Vector

**Circulation:** The line integral of a continuous vector function $\mathbf{F}$ along a closed smooth curve $C$ is called the circulation of $\mathbf{F}$ along $C$ and is sometimes denoted by

$$\oint_C \mathbf{F} \cdot d\mathbf{r} \tag{47}$$

**Simply connected region:** A region (or domain) $D$ is called simply connected if every closed curve in $D$ can be continuously shrunk to any point in $D$ without leaving $D$.

**Irrotational vector:** A continuous vector function $\mathbf{F}$ is said to be irrotational in a simply connected region $D$, if its circulation along every closed curve in $D$ vanishes, i.e.,

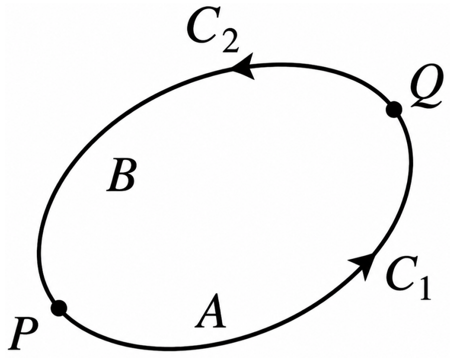
$$\oint_C \mathbf{F} \cdot d\mathbf{r} = 0 \tag{48}$$

**Theorem 4.1.2.** *The line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ of a continuous vector function $\mathbf{F}$ defined in a region $D$ of space is independent of path in $D$ if and only if*

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = 0 \tag{49}$$

*on every simple closed path in $D$.*

*Proof.* Let the line integral be independent of path in $D$.




Let $C$ be a simple closed path in $D$. Subdivide $C$ into two paths $C_1 = PAQ$ and $C_2 = QBP$. Then

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \int_{C_1} \mathbf{F} \cdot d\mathbf{r} + \int_{C_2} \mathbf{F} \cdot d\mathbf{r} = \int_{PAQ} \mathbf{F} \cdot d\mathbf{r} + \int_{QBP} \mathbf{F} \cdot d\mathbf{r}$$

Since the integral is independent of path from $P$ to $Q$,

$$\int_{PAQ} \mathbf{F} \cdot d\mathbf{r} = \int_{PBQ} \mathbf{F} \cdot d\mathbf{r} = -\int_{QBP} \mathbf{F} \cdot d\mathbf{r}$$

Hence,

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = -\int_{QBP} \mathbf{F} \cdot d\mathbf{r} + \int_{QBP} \mathbf{F} \cdot d\mathbf{r} = 0$$

Conversely, let the integral be zero on every simple closed path in $D$. Let $P$ and $Q$ be two points in $D$ and let $C_1 = PAQ$ and $C_2 = QBP$ be two paths joining them. Then they form a closed path $C$, and

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = 0$$

or

$$\int_{PAQ} \mathbf{F} \cdot d\mathbf{r} + \int_{QBP} \mathbf{F} \cdot d\mathbf{r} = 0$$

or

$$\int_{PAQ} \mathbf{F} \cdot d\mathbf{r} = -\int_{QBP} \mathbf{F} \cdot d\mathbf{r} = \int_{PBQ} \mathbf{F} \cdot d\mathbf{r}$$

which shows that the value of the integral from $P$ to $Q$ is the same for both paths. Hence the integral is independent of path. □

**Theorem 4.1.3.** *The line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ of a continuous vector function $\mathbf{F}$ having continuous first partial derivatives in a simply connected region $D$ is independent of path in $D$ if and only if*

$$\nabla \times \mathbf{F} = 0 \tag{50}$$

*everywhere in $D$.*

*Proof.* If the line integral is independent of path in $D$, then by Theorem 1 there exists a scalar function $\phi$ such that $\mathbf{F} = \nabla\phi$. Hence,

$$\nabla \times \mathbf{F} = \nabla \times (\nabla\phi) = 0$$

Conversely, if $\nabla \times \mathbf{F} = 0$ everywhere in a simply connected region, then $\mathbf{F}$ is conservative and hence the line integral is independent of path.

Let $C$ be a simple closed path in $D$. Since $D$ is simply connected, we can find a surface $S$ in $D$ bounded by $C$. Then by Stokes' theorem, we have

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \hat{n}\, dS = 0$$

Hence, by Theorem 2, the integral is independent of path. □

**Note:** From above, we conclude that under appropriate conditions, the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of path if and only if

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = 0 \quad \text{or} \quad \mathbf{F} = \nabla\phi \quad \text{or} \quad \nabla \times \mathbf{F} = 0.$$

The field $\mathbf{F}$ of this character is called a *conservative field* and $\phi$ is called the *potential* of this field.

## 4.2 Surface Integral

Consider a simple, smooth and orientable surface $S$. The surface $S$ is *simple* in the sense that it has no points at which it intersects or touches itself; it is *smooth* in the sense that it has a unique normal at every point of it; and it is *orientable* in the sense that a chosen positive normal direction at any point of $S$ can be continued in a unique and continuous manner to the entire surface.

We consider here only two-sided surfaces which are orientable. Surfaces such as the Möbius strip or Klein bottle are excluded. The positive normal will be assumed conventionally to be in the outward direction for a closed surface and in a right-handed sense for an open surface bounded by a closed curve $C$.

Let a continuous vector function $\mathbf{F}$ be defined at each point of $S$. Let $S$ be arbitrarily subdivided into $n$ parts $S_1, S_2, \ldots, S_n$ of areas $\delta S_1, \delta S_2, \ldots, \delta S_n$. Let $Q_1, Q_2, \ldots, Q_n$ be arbitrary points chosen in these parts, and let $\mathbf{F}(Q_i)$ be the value of $\mathbf{F}$ at $Q_i$, and let $\hat{n}_i$ be the positive (outward) unit normal vector to $S_i$ at $Q_i$, $i = 1, 2, \ldots, n$.

Form the sum,

$$I_n = \sum_{i=1}^{n} \mathbf{F}(Q_i) \cdot \hat{n}_i\, \delta S_i$$

and find the limit of this sum as $n \to \infty$ and every $\delta S_i \to 0$. Since $\mathbf{F}$ is continuous and $S$ is smooth, this limit exists and is independent of the mode of subdivision of $S$ and the points $Q_i$.

This limit is called the *surface integral* of $\mathbf{F}$ over $S$ and is denoted by

$$\iint_S \mathbf{F} \cdot \hat{n}\, dS \tag{51}$$

### 4.2.1 A Evaluation of Surface Integrals

Normally the surface integrals are evaluated by reducing them to double integrals over a plane region, as follows.

**Theorem 4.2.1.** *If $R_1, R_2$ and $R_3$ be the orthogonal projections of $S$ in $yz$-plane, $zx$-plane and $xy$-plane respectively, then*

$$\iint_S \mathbf{F}\cdot\hat{n}\,dS = \iint_{R_1} \mathbf{F}\cdot\hat{n}\,\frac{dy\,dz}{|\hat{n}\cdot\mathbf{i}|} = \iint_{R_2} \mathbf{F}\cdot\hat{n}\,\frac{dz\,dx}{|\hat{n}\cdot\mathbf{j}|} = \iint_{R_3} \mathbf{F}\cdot\hat{n}\,\frac{dx\,dy}{|\hat{n}\cdot\mathbf{k}|} \tag{52}$$

**Proof:** If $\alpha, \beta, \gamma$ be the direction angles of $\hat{n}$, then

$$\hat{n} = \mathbf{i}\cos\alpha + \mathbf{j}\cos\beta + \mathbf{k}\cos\gamma$$

and the orthogonal projections of $dS$ on $yz$, $zx$ and $xy$ planes are respectively

$$dS\cos\alpha, \quad dS\cos\beta, \quad dS\cos\gamma.$$

Then

$$dy\,dz = dS\cos\alpha, \quad dz\,dx = dS\cos\beta, \quad dx\,dy = dS\cos\gamma$$

or

$$dy\,dz = dS\,|\hat{n}\cdot\mathbf{i}|, \quad dz\,dx = dS\,|\hat{n}\cdot\mathbf{j}|, \quad dx\,dy = dS\,|\hat{n}\cdot\mathbf{k}|$$

Hence,

$$dS = \frac{dy\,dz}{|\hat{n}\cdot\mathbf{i}|} = \frac{dz\,dx}{|\hat{n}\cdot\mathbf{j}|} = \frac{dx\,dy}{|\hat{n}\cdot\mathbf{k}|}$$

Substituting, we obtain (52).

**Note:** Since projected area is always taken positive, $dS\cos\alpha$ will be positive when $\alpha$ is acute and negative when $\alpha$ is obtuse. Hence we must take

$$dy\,dz = dS|\cos\alpha| = dS|\hat{n}\cdot\mathbf{i}|, \quad \text{etc.}$$

### 4.2.2 Volume integral

Volume integral of a continuous vector function $\mathbf{F}$ is defined by

$$\iiint_V \mathbf{F}\,dV \tag{53}$$

**Example 4.1** If the acceleration of a particle at time $t$ is given by

$$\mathbf{a} = e^{-t}\mathbf{i} - 6(t+1)\mathbf{j} + 3\sin t\,\mathbf{k},$$

find the velocity $\mathbf{v}$ and displacement $\mathbf{r}$ at time $t$, given that $\mathbf{v} = 0$ and $\mathbf{r} = 0$ at $t = 0$.

**Solution.**

$$\begin{aligned}\mathbf{v} = \int \mathbf{a}\,dt &= \int \left(e^{-t}\mathbf{i} - 6(t+1)\mathbf{j} + 3\sin t\,\mathbf{k}\right) dt \\ &= e^{-t}\mathbf{i} - (3t^2+6t)\mathbf{j} - 3\cos t\,\mathbf{k} + \mathbf{c}_1\end{aligned}$$

At $t = 0$, $\mathbf{v} = 0$:

$$0 = \mathbf{i} - 3\mathbf{k} + \mathbf{c}_1 \Rightarrow \mathbf{c}_1 = -\mathbf{i} + 3\mathbf{k}$$

Hence,

$$\mathbf{v} = (1 - e^{-t})\mathbf{i} - (3t^2 + 6t)\mathbf{j} + (3 - 3\cos t)\mathbf{k}$$

Now,

$$\begin{aligned}\mathbf{r} &= \int \mathbf{v}\,dt \\ &= \int \left[(1 - e^{-t})\mathbf{i} - (3t^2 + 6t)\mathbf{j} + (3 - 3\cos t)\mathbf{k}\right] dt \\ &= (t - e^{-t})\mathbf{i} - (t^3 + 3t^2)\mathbf{j} + (3t - 3\sin t)\mathbf{k} + \mathbf{c}_2\end{aligned}$$

At $t = 0$, $\mathbf{r} = 0$:

$$0 = -\mathbf{i} + \mathbf{c}_2 \Rightarrow \mathbf{c}_2 = \mathbf{i}$$

Hence,

$$\mathbf{r} = (t - 1 + e^{-t})\mathbf{i} - (t^3 + 3t^2)\mathbf{j} + (3t - 3\sin t)\mathbf{k}$$

**Example 4.2** Evaluate the line integral $\int_C \mathbf{F}\cdot d\mathbf{r}$, where

$$\mathbf{F} = (x+y)\mathbf{i} + (2x - z)\mathbf{j} + (y+z)\mathbf{k}$$

and $C$ is the boundary of the triangle with vertices $(2,0,0)$, $(0,3,0)$ and $(0,0,6)$ and orientation $(2,0,0) \to (0,3,0) \to (0,0,6)$.

**Solution.** We have

$$\begin{aligned}\mathbf{F}\cdot d\mathbf{r} &= (x+y)\,dx + (2x-z)\,dy + (y+z)\,dz \\ \Rightarrow \quad \int_C \mathbf{F}\cdot d\mathbf{r} &= \int_{AB} \mathbf{F}\cdot d\mathbf{r} + \int_{BC} \mathbf{F}\cdot d\mathbf{r} + \int_{CA} \mathbf{F}\cdot d\mathbf{r}\end{aligned}$$

**For $AB$:** Parametric equations are

$$\frac{x-2}{0-2} = \frac{y-0}{3-0} = \frac{z-0}{0-0} = t$$
$$x = 2 - 2t, \quad y = 3t, \quad z = 0$$

with $t = 0$ at $A$ and $t = 1$ at $B$.

$$\begin{aligned}\int_{AB} \mathbf{F}\cdot d\mathbf{r} &= \int_0^1 [(2 - 2t + 3t)(-2dt) + (4 - 4t - 0)(3dt) + (3t + 0)0] \\ &= \int_0^1 (8 - 14t)\,dt = (8t - 7t^2)_0^1 = 1\end{aligned}$$

**For $BC$:**

$$\frac{x-0}{0-0} = \frac{y-3}{0-3} = \frac{z-0}{6-0} = t$$
$$x = 0, \quad y = 3-3t, \quad z = 6t$$

$$\begin{aligned}\int_{BC} \mathbf{F}\cdot d\mathbf{r} &= \int_0^1 [(2+3-3t)0 + (0-6t)(-3dt) + (3-3t+6t)(6dt)] \\ &= \int_0^1 (36t+18t)\,dt = (18t^2+18t)_0^1 = 36\end{aligned}$$

**For $CA$:**

$$\frac{x-0}{2-0} = \frac{y-0}{0-0} = \frac{z-6}{0-6} = t$$
$$x = 2t, \quad y = 0, \quad z = 6-6t$$

$$\begin{aligned}\int_{CA} \mathbf{F}\cdot d\mathbf{r} &= \int_0^1 [(2t)(2dt) + (4t-6+6t)0 + (0+6-6t)(-6dt)] \\ &= \int_0^1 (40t-36)\,dt = (20t^2-36t)_0^1 = -16\end{aligned}$$

Hence,

$$\int_C \mathbf{F}\cdot d\mathbf{r} = 1+36-16 = 21$$

**Example 4.3** Compute $\int_C \mathbf{F}\cdot d\mathbf{r}$, where

$$\mathbf{F} = \frac{-y\mathbf{i}+x\mathbf{j}}{x^2+y^2}$$

and $C$ is the circular path $x^2+y^2=a^2$ described in the counterclockwise sense.
**Solution.** Let

$$x = a\cos\theta, \quad y = a\sin\theta \quad (0 \le \theta \le 2\pi)$$
$$dx = -a\sin\theta\,d\theta, \quad dy = a\cos\theta\,d\theta$$

Then,

$$\begin{aligned}\int_C \mathbf{F}\cdot d\mathbf{r} &= \int \frac{-y\,dx + x\,dy}{x^2+y^2} \\ &= \int_0^{2\pi} \frac{(-a\sin\theta)(-a\sin\theta)+(a\cos\theta)(a\cos\theta)}{a^2} d\theta \\ &= \int_0^{2\pi} d\theta = 2\pi\end{aligned}$$

**Comment:** It is possible to write $\mathbf{F} = \nabla\phi$, where $\phi = \tan^{-1}(y/x)$ so that $\nabla\times\mathbf{F} = 0$ except at the origin. This means $\mathbf{F}$ is not conservative over the whole space including the origin, and hence the integral around a closed curve is not necessarily zero.

**Example 4.4** Let

$$\mathbf{F} = yz^2\mathbf{i} + xz^2\mathbf{j} + 2xyz\mathbf{k}$$

and $C$ be the path defined by $x^2+y^2+z^2=1$, $x=y$, from $(1/\sqrt{2}, 1/\sqrt{2}, 0)$ to $(1/2, 1/2, \sqrt{2}/2)$. Find $\int_C \mathbf{F}\cdot d\mathbf{r}$.

**Solution.** We have

$$\nabla\times\mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ yz^2 & xz^2 & 2xyz \end{vmatrix} = (2xz-2xz)\mathbf{i} + (2yz-2yz)\mathbf{j} + (z^2-z^2)\mathbf{k} = 0$$

Since $\nabla\times\mathbf{F}=0$, we have $\mathbf{F}=\nabla\phi$, such that

$$\frac{\partial\phi}{\partial x} = yz^2, \quad \frac{\partial\phi}{\partial y} = xz^2, \quad \frac{\partial\phi}{\partial z} = 2xyz$$

**Example 4.5** Evaluate $\iint_S \mathbf{F}\cdot\hat{n}\,dS$, where $\mathbf{F} = x\hat{\imath}+y\hat{\jmath}+z\hat{k}$ and $S$ is the surface of the unit cube bounded by the coordinate planes and the planes $x=1$, $y=1$, $z=1$.

**Solution.**

We have six faces $S_1, S_2, S_3, S_4, S_5, S_6$.

$$\begin{aligned}
&\text{On } S_1: x=0,\ \hat{n}=-\hat{\imath},\ dS=dydz \\
&\text{On } S_2: x=1,\ \hat{n}=\hat{\imath},\ dS=dydz \\
&\text{On } S_3: y=0,\ \hat{n}=-\hat{\jmath},\ dS=dzdx \\
&\text{On } S_4: y=1,\ \hat{n}=\hat{\jmath},\ dS=dzdx \\
&\text{On } S_5: z=0,\ \hat{n}=-\hat{k},\ dS=dxdy \\
&\text{On } S_6: z=1,\ \hat{n}=\hat{k},\ dS=dxdy
\end{aligned}$$

Now,

$$\iint_{S_1} \mathbf{F}\cdot\hat{n}\,dS = \int_0^1\int_0^1 (x\hat{\imath}+y\hat{\jmath}+z\hat{k})\cdot(-\hat{\imath})\,dydz = \int_0^1\int_0^1 (-x)\,dydz = 0$$

Similarly,

$$\begin{aligned}
\iint_{S_2} \mathbf{F}\cdot\hat{n}\,dS &= \int_0^1\int_0^1 x\,dydz = 1 \\
\iint_{S_3} \mathbf{F}\cdot\hat{n}\,dS &= \int_0^1\int_0^1 0\,dzdx = 0 \\
\iint_{S_4} \mathbf{F}\cdot\hat{n}\,dS &= \int_0^1\int_0^1 y\,dzdx = 1 \\
\iint_{S_5} \mathbf{F}\cdot\hat{n}\,dS &= 0 \\
\iint_{S_6} \mathbf{F}\cdot\hat{n}\,dS &= 1
\end{aligned}$$

Therefore,

$$\iint_S \mathbf{F}\cdot\hat{n}\,dS = 0+1+0+1+0+1 = 3$$

**Example 4.6** Evaluate $\iint_S (yz\hat{i} + zx\hat{j} + xy\hat{k}) \cdot d\mathbf{S}$, where $S$ is the surface of the sphere $x^2 + y^2 + z^2 = a^2$.

**Solution.**

Let

$$\mathbf{F} = yz\hat{i} + zx\hat{j} + xy\hat{k}$$

The surface is given by:

$$\phi = x^2 + y^2 + z^2 - a^2 = 0$$

$$\hat{n} = \frac{\nabla\phi}{|\nabla\phi|} = \frac{2x\hat{i} + 2y\hat{j} + 2z\hat{k}}{\sqrt{4x^2 + 4y^2 + 4z^2}} = \frac{x\hat{i} + y\hat{j} + z\hat{k}}{a}$$

Thus,

$$\mathbf{F} \cdot \hat{n} = \frac{xyz + xyz + xyz}{a} = \frac{3xyz}{a}$$

Using spherical coordinates:

$$x = a\sin\theta\cos\phi, \quad y = a\sin\theta\sin\phi, \quad z = a\cos\theta$$

So,

$$xyz = a^3\sin^2\theta\cos\theta\sin\phi\cos\phi$$

Surface element:

$$dS = a^2\sin\theta\, d\theta\, d\phi$$

Hence,

$$\iint_S \mathbf{F} \cdot \hat{n}\, dS = \int_0^{2\pi}\int_0^{\pi} 3a^4\sin^3\theta\cos\theta\sin\phi\cos\phi\, d\theta\, d\phi$$

Separating integrals:

$$= 3a^4\left(\int_0^{\pi}\sin^3\theta\cos\theta\, d\theta\right)\left(\int_0^{2\pi}\sin\phi\cos\phi\, d\phi\right)$$

$$\int_0^{2\pi}\sin\phi\cos\phi\, d\phi = 0$$

Therefore,

$$\iint_S \mathbf{F} \cdot \hat{n}\, dS = 0$$

**Example 4.7** Evaluate

$$\iint_S (x^3\, dydz + y^3\, dzdx + z^3\, dxdy)$$

where $S$ is the closed surface consisting of the cylinder $x^2 + y^2 = a^2$, $0 \le z \le b$, and the planes $z = 0$ and $z = b$.

We have three surfaces: curved surface $S_1$, plane $S_2$, and plane $S_3$. The integral is:

$$\iint_S \mathbf{F} \cdot \hat{n}\, dS, \quad \text{where } \mathbf{F} = x^3\hat{i} + x^2y\hat{j} + x^2z\hat{k}.$$

**On $S_1$ (curved surface)**

$$x^2+y^2=a^2, \quad x=a\cos\phi,\ y=a\sin\phi,\ z=z$$

$$\hat{n}=\frac{\nabla f}{|\nabla f|}=\frac{x\hat{i}+y\hat{j}}{a}, \quad dS=a\,d\phi\,dz$$

$$\mathbf{F}\cdot\hat{n}=\frac{x^4+x^2y^2}{a}=a^3\cos^2\phi$$

$$\iint_{S_1}\mathbf{F}\cdot\hat{n}\,dS=\int_0^b\int_0^{2\pi}a^4\cos^2\phi\,d\phi\,dz=\pi a^4 b$$

**On $S_2$ (bottom plane $z=0$)**

$$x^2+y^2\le a^2, \quad \hat{n}=-\hat{k}, \quad dS=dxdy$$

$$\iint_{S_2}\mathbf{F}\cdot\hat{n}\,dS=\iint -x^2z\,dxdy=0$$

**On $S_3$ (top plane $z=b$)**

$$x^2+y^2\le a^2, \quad \hat{n}=\hat{k}, \quad dS=dxdy$$

$$\iint_{S_3}\mathbf{F}\cdot\hat{n}\,dS=\iint x^2b\,dxdy$$

Convert to polar:

$$=b\int_0^{2\pi}\int_0^a r^2\cos^2\theta\,r\,dr\,d\theta=b\int_0^{2\pi}\cos^2\theta\,d\theta\int_0^a r^3dr$$

$$=b\cdot\pi\cdot\frac{a^4}{4}=\frac{\pi a^4 b}{4}$$

**Total**

$$\iint_S\mathbf{F}\cdot\hat{n}\,dS=\pi a^4b+0+\frac{\pi a^4 b}{4}=\frac{5\pi a^4 b}{4}$$

**Example 4.8** Evaluate $\iint_S \mathbf{F}\cdot\hat{n}\,dS$, where $\mathbf{F}=x^2\hat{i}+y^2\hat{j}+z^2\hat{k}$ and $S$ is the portion of the plane $x+y+z=a$ in the first octant.

**Solution.**

Plane:

$$\phi=x+y+z-a=0$$

$$\hat{n}=\frac{\nabla\phi}{|\nabla\phi|}=\frac{\hat{i}+\hat{j}+\hat{k}}{\sqrt{3}}$$

$$\mathbf{F}\cdot\hat{n} = \frac{x^2+y^2+z^2}{\sqrt{3}} = \frac{x^2+y^2+(a-x-y)^2}{\sqrt{3}}$$

Also,

$$dS = \frac{\sqrt{3}}{\hat{n}\cdot\hat{k}}\,dxdy = \sqrt{3}\,dxdy$$

Thus,

$$\iint_S \mathbf{F}\cdot\hat{n}\,dS = \iint_R \left[x^2+y^2+(a-x-y)^2\right]dxdy$$

Region:

$$R: x\geq 0,\ y\geq 0,\ x+y\leq a$$

$$= \int_0^a\int_0^{a-x}\left[x^2+y^2+(a-x-y)^2\right]dy\,dx$$

After integration:

$$= \frac{a^4}{4}$$

**Example 4.9** Evaluate

$$\iiint_V (ax^2+by^2+cz^2)\,dx\,dy\,dz$$

where $V$ is the volume of the sphere:

$$x^2+y^2+z^2\leq 1.$$

**Solution.**

The region is bounded by:

$$z = \pm\sqrt{1-x^2-y^2}$$

Thus,

$$\iiint_V (ax^2+by^2+cz^2)\,dx\,dy\,dz = \iint_R\int_{-\sqrt{1-x^2-y^2}}^{\sqrt{1-x^2-y^2}}(ax^2+by^2+cz^2)\,dz\,dx\,dy$$

$$= 2\iint_R\left[(ax^2+by^2)\sqrt{1-x^2-y^2}+\frac{c}{3}(1-x^2-y^2)^{3/2}\right]dxdy$$

Switch to polar:

$$x = r\cos\theta,\quad y = r\sin\theta,\quad R: r\leq 1$$

Using polar coordinates $x=\rho\cos\phi$, $y=\rho\sin\phi$, with $0\leq\rho\leq 1$, $0\leq\phi\leq 2\pi$ and $dA=\rho\,d\rho\,d\phi$:

$$\begin{aligned}
&2\int_0^1\int_0^{2\pi}\sqrt{1-\rho^2}\left[a\rho^2\cos^2\phi+b\rho^2\sin^2\phi+\frac{c}{3}(1-\rho^2)\right]\rho\,d\phi\,d\rho\\
&= 2\int_0^1\rho\sqrt{1-\rho^2}\left[\int_0^{2\pi}(a\rho^2\cos^2\phi+b\rho^2\sin^2\phi+\frac{c}{3}(1-\rho^2))d\phi\right]d\rho\\
&= 2\pi\int_0^1\rho\sqrt{1-\rho^2}\left[a\rho^2+b\rho^2+\frac{2c}{3}(1-\rho^2)\right]d\rho\\
&= \frac{4\pi}{15}(a+b+c)
\end{aligned}$$

**Example 4.10** Show that

$$\mathbf{F} = (2xy + z^3)\hat{\imath} + x^2\hat{\jmath} + 3xz^2\hat{k}$$

is a conservative force field. Find the potential and calculate the work done in moving a particle from $(1, -2, 1)$ to $(3, 1, 4)$.

**Solution.**

$$\nabla \times \mathbf{F} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xy + z^3 & x^2 & 3xz^2 \end{vmatrix} = 0$$

Hence $\mathbf{F}$ is conservative, so $\mathbf{F} = \nabla\phi$.

$$\phi = \int \mathbf{F} \cdot d\mathbf{r} = \int (F_1 dx + F_2 dy + F_3 dz)$$

Choose path in segments:

$$\phi = \int_{x_0}^{x} (2xy_0 + z_0^3) dx + \int_{y_0}^{y} x^2 dy + \int_{z_0}^{z} 3xz^2 dz$$

$$= (x^2 y_0 + xz_0^3)\Big|_{x_0}^{x} + (x^2 y)\Big|_{y_0}^{y} + (xz^3)\Big|_{z_0}^{z}$$

Thus,

$$\phi = x^2 y + xz^3$$

Work done:

$$W = \phi(3, 1, 4) - \phi(1, -2, 1)$$

$$= [3^2 \cdot 1 + 3 \cdot 4^3] - [1^2(-2) + 1 \cdot 1^3] = (9 + 192) - (-2 + 1) = 202$$

## 4.3 Exercises

1. Evaluate the volume integral $\iiint_V xyz\, dx\, dy\, dz$ over a domain bounded by $x = 0$, $y = 0$, $z = 0$, $x + y + z = 1$.

2. Evaluate $\iiint_V z^2\, dx\, dy\, dz$, where $V$ is the volume of the ellipsoid

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \leq 1.$$

3. Calculate $\iiint_V z^2\, dx\, dy\, dz$, where $V$ is the volume in the first octant $(x > 0, y > 0, z > 0)$ bounded by the plane $x + y + z = a$.

4. Evaluate $\iint_S (\nabla \times \mathbf{F}) \cdot \hat{n}\, dS$, where $\mathbf{F} = y^2\hat{\imath} + x^2\hat{\jmath} + xyz\hat{k}$ and $S$ is the hemisphere $x^2 + y^2 + z^2 = a^2$, $z \geq 0$.

5. Calculate $\iint_S xyz\, dS$, where $S$ is the part of the plane $ax + by + cz = d$ which lies in the first octant.

6. Evaluate $\oint_C \mathbf{F} \cdot d\mathbf{r}$, where $\mathbf{F} = 3x^2y \sin z\,\hat{\imath} + 2x^3y \sin z\,\hat{\jmath} + x^3y^2 \cos z\,\hat{k}$ and $C$ is a closed contour.

7. Evaluate $\iiint_V (x^2 + y^2 + z^2)^{3/2}\, dx\, dy\, dz$ over the volume $x^2 + y^2 + z^2 \leq a^2$.

## 4.4 Green's Theorem in the Plane

**Theorem 4.4.1.** *If $R$ is a closed bounded region in the $xy$-plane with boundary $C$ consisting of finitely many smooth curves and $F_1(x, y)$ and $F_2(x, y)$ are functions which are continuous and have continuous partial derivatives $\frac{\partial F_1}{\partial y}$ and $\frac{\partial F_2}{\partial x}$ everywhere in some domain containing $R$, then*

$$\iint_R \left( \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) dx\, dy = \oint_C (F_1\, dx + F_2\, dy). \tag{54}$$

*where the line integral is taken in the positive direction, i.e., $R$ lies on the left as one advances in the direction of integration.*

**Note:**

1. If $\dfrac{\partial F_2}{\partial x} = \dfrac{\partial F_1}{\partial y}$, then $F_1 dx + F_2 dy$ is an exact differential.

2. If $\dfrac{\partial F_2}{\partial x} = \dfrac{\partial F_1}{\partial y}$, then the line integral $\displaystyle\int_A^B (F_1 dx + F_2 dy)$ is independent of path from $A$ to $B$.

3. If $\int_A^B (F_1 dx + F_2 dy)$ is independent of path from $A$ to $B$, then $\dfrac{\partial F_2}{\partial x} = \dfrac{\partial F_1}{\partial y}$.

## 4.5 Gauss's Divergence Theorem

**Theorem 4.5.1.** *If $V$ be a closed bounded region in space whose boundary is a piecewise smooth orientable surface $S$ and $\mathbf{F}(x, y, z)$ be a vector function which is continuous and has continuous first partial derivatives in some domain containing $V$, then the volume integral of the divergence of $\mathbf{F}$ taken over the volume $V$ enclosed by the surface $S$ is equal to the surface integral of the normal component of $\mathbf{F}$ taken over the surface $S$, i.e.,*

$$\iiint_V \operatorname{div} \mathbf{F}\, dV = \iint_S \mathbf{F} \cdot \hat{n}\, dS \tag{55}$$

*where $\hat{n}$ is the outward unit normal vector to $S$.*

## 4.6 Stokes's Theorem

**Theorem 4.6.1.** *If $S$ be a piecewise smooth oriented surface whose boundary is a piecewise smooth simple closed curve $C$ and if $\mathbf{F}(x, y, z)$ be a vector function which is continuous and has continuous first partial derivatives in a region in space containing $S$, then the surface integral of the component of curl $\mathbf{F}$ normal to the surface $S$ is equal to the line integral of $\mathbf{F}$ taken round the curve $C$, the normal to $S$ being taken in the right-handed sense with respect to orientation of $C$, i.e.*

$$\iint_S (\operatorname{curl} \mathbf{F}) \cdot \hat{n}\, dS = \oint_C \mathbf{F} \cdot d\mathbf{r} \tag{56}$$

### 4.6.1 Green's theorem in the plane as a special case of Stokes' theorem

Let $\mathbf{F}(x, y) = F_1(x, y)\mathbf{i} + F_2(x, y)\mathbf{j}$ be a vector function which is continuously differentiable in a simply connected bounded closed region $R$ whose boundary is a piecewise smooth closed curve $C$. Then $dS = dx\, dy$ and $d\mathbf{r} = dx\, \mathbf{i} + dy\, \mathbf{j}$.

Now,

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & 0 \end{vmatrix} = \left( \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) \mathbf{k}$$

Thus,

$$\hat{n} \cdot (\nabla \times \mathbf{F}) = \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \quad (\text{if } \hat{n} \parallel \mathbf{k})$$

Also,

$$\mathbf{F} \cdot d\mathbf{r} = (F_1 \mathbf{i} + F_2 \mathbf{j}) \cdot (dx\, \mathbf{i} + dy\, \mathbf{j}) = F_1 dx + F_2 dy$$

Hence, Stokes' theorem becomes

$$\iint_R \left( \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) dx\, dy = \oint_C (F_1 dx + F_2 dy) \tag{57}$$

which is Green's theorem in the plane.

**Example 4.11** Using Green's theorem, evaluate

$$\oint_C [2xy\, dx + (e^x + x^2)\, dy]$$

where $C$ is the boundary of the triangle with vertices $(0, 0)$, $(1, 0)$, $(1, 1)$ (clockwise).

Since curve $C : OBAO$ is described in the clockwise sense, Green's theorem becomes

$$\oint_C (F_1 dx + F_2 dy) = \iint_R \left( \frac{\partial F_1}{\partial y} - \frac{\partial F_2}{\partial x} \right) dx\, dy$$

Here,

$$\begin{aligned}
F_1 &= 2xy, \quad F_2 = e^x + x^2 \\
\frac{\partial F_1}{\partial y} &= 2x, \quad \frac{\partial F_2}{\partial x} = e^x + 2x
\end{aligned}$$

$$\begin{aligned}
\Rightarrow \oint_C [2xy\,dx + (e^x + x^2)\,dy] &= \iint_R [2x - (e^x + 2x)]\,dx\,dy = \iint_R (-e^x)\,dx\,dy \\
&= \int_0^1 \int_0^x (-e^x)\,dy\,dx = \int_0^1 (-xe^x)\,dx \\
&= (-xe^x + e^x)\Big|_0^1 = -1
\end{aligned}$$

**Example 4.12** Verify Gauss's divergence theorem for the vector function

$$\mathbf{F} = x^2\mathbf{i} + y^2\mathbf{j} + z^2\mathbf{k}$$

and the region bounded by the cylinder $x^2 + y^2 = k^2$ and the planes $z = h$ and $z = -h$.
**Solution.** The divergence theorem states that

$$\iiint_V (\nabla \cdot \mathbf{F})dV = \iint_S \mathbf{F} \cdot \hat{n} dS$$

(i) Since $\mathbf{F} = x^2\mathbf{i} + y^2\mathbf{j} + z^2\mathbf{k}$,

$$\nabla \cdot \mathbf{F} = 2x + 2y + 2z$$

$$\therefore \iiint_V (\nabla \cdot \mathbf{F})dV = \iiint_V 2(x + y + z)\,dV$$

Transforming to cylindrical coordinates $(\rho, \phi, z)$:

$$x = \rho\cos\phi, \quad y = \rho\sin\phi, \quad z = z$$

$$0 \le \rho \le k, \quad 0 \le \phi \le 2\pi, \quad -h \le z \le h$$

$$dV = \rho\,d\rho\,d\phi\,dz$$

$$\iiint_V (\nabla \cdot \mathbf{F})dV = \int_{-h}^{h} \int_0^{2\pi} \int_0^k 2(\rho\cos\phi + \rho\sin\phi + z)\rho\,d\rho\,d\phi\,dz$$

$$= 4h \int_0^k \rho^2 d\rho \int_0^{2\pi} (\cos\phi + \sin\phi)\,d\phi = 0$$

(ii) Now evaluate surface integral:

$$\iint_S \mathbf{F} \cdot \hat{n} dS = \iint_{S_1} + \iint_{S_2} + \iint_{S_3}$$

**On** $S_1$ (curved surface): $x^2 + y^2 = k^2$

$$\hat{n} = \frac{x\mathbf{i} + y\mathbf{j}}{k}, \quad x = k\cos\phi, \quad y = k\sin\phi, \quad dS = k\,d\phi\,dz$$

$$\mathbf{F} \cdot \hat{n} = \frac{x^3 + y^3}{k} = k^2(\cos^3\phi + \sin^3\phi)$$

$$I_1 = \int_{-h}^{h} \int_0^{2\pi} k^2(\cos^3\phi + \sin^3\phi) k\,d\phi\,dz = 0$$

**On** $S_2$ (top surface $z = h$):

$$\hat{n} = \mathbf{k}, \quad \mathbf{F} \cdot \hat{n} = z^2 = h^2$$

$$I_2 = \iint h^2\,dx\,dy = h^2 \cdot \pi k^2$$

**On** $S_3$ (bottom surface $z = -h$):

$$\hat{n} = -\mathbf{k}, \quad \mathbf{F} \cdot \hat{n} = -z^2 = -h^2$$

$$I_3 = -h^2 \cdot \pi k^2$$

$$\therefore \iint_S \mathbf{F} \cdot \hat{n} dS = 0$$

Hence verified.
On $S_3$: $\hat{n} = -\mathbf{k}, \quad \mathbf{F} \cdot \hat{n} = -z^2 = -h^2, \quad dS = dx\,dy$

$$I_3 = \int_{-k}^{k} \int_{-\sqrt{k^2-x^2}}^{\sqrt{k^2-x^2}} (-h^2)\,dy\,dx = -\pi h^2 k^2$$

$$\therefore \iint_S \mathbf{F} \cdot \hat{n} dS = 0 + \pi h^2 k^2 - \pi h^2 k^2 = 0$$

**Example 4.13** Verify Stokes' theorem for the vector field

$$\mathbf{F} = (y^2 + z^2 - x^2)\mathbf{i} + (z^2 + x^2 - y^2)\mathbf{j} + (x^2 + y^2 - z^2)\mathbf{k}$$

over the portion of the surface

$$x^2 + y^2 + z^2 - 2ax - 2cz = 0, \quad z \geq 0$$

**Solution.** Stokes' theorem states:

$$\iint_S (\nabla \times \mathbf{F}) \cdot \hat{n}\,dS = \oint_C \mathbf{F} \cdot d\mathbf{r}$$

**(i)** Consider orthogonal projection of $S$ on the $xy$-plane, which is the circular area

$$R : x^2 + y^2 - 2ax = 0, \quad z = 0$$

with centre at $(a, 0)$ and radius $a$.

Then,

$$\iint_S (\nabla \times \mathbf{F}) \cdot \hat{n}\, dS = \iint_R (\nabla \times \mathbf{F}) \cdot \hat{n}\, dR$$

Now,

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2 + z^2 - x^2 & z^2 + x^2 - y^2 & x^2 + y^2 - z^2 \end{vmatrix}$$

$$= 2(y - z)\mathbf{i} + 2(z - x)\mathbf{j} + 2(x - y)\mathbf{k}$$

On $R$: $\hat{n} = \mathbf{k}$, $dR = r\, dr\, d\theta$,

$$x = a + r\cos\theta, \quad y = r\sin\theta, \quad z = 0$$

$$0 \le r \le a, \quad 0 \le \theta \le 2\pi$$

Thus,

$$(\nabla \times \mathbf{F}) \cdot \hat{n} = 2(x - y) = 2(a + r\cos\theta - r\sin\theta)$$

$$\iint_S (\nabla \times \mathbf{F}) \cdot \hat{n}\, dS = \int_0^{2\pi} \int_0^a 2(a + r\cos\theta - r\sin\theta) r\, dr\, d\theta$$

$$= \frac{2a^3}{3} \int_0^{2\pi} \left( \frac{3}{2} + \cos\theta - \sin\theta \right) d\theta = 2\pi a^3$$

**(ii)** On $C$: $x = a + a\cos\theta$, $y = a\sin\theta$, $z = 0$

$$\mathbf{F} \cdot d\mathbf{r} = (y^2 + z^2 - x^2)dx + (z^2 + x^2 - y^2)dy + (x^2 + y^2 - z^2)dz$$

$$= 2a^3(\sin\theta + \cos\theta)\cos\theta(1 + \cos\theta)\, d\theta$$

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \int_0^{2\pi} 2a^3(\sin\theta + \cos\theta)\cos\theta(1 + \cos\theta)\, d\theta = 2\pi a^3$$

Hence, the theorem is verified.

**Example 4.14** Evaluate $\oint_C \mathbf{F} \cdot d\mathbf{r}$ by Stokes' theorem, where

$$\mathbf{F} = y\mathbf{i} + xz^2\mathbf{j} - zy^3\mathbf{k}$$

and $C$ is the circle $x^2 + y^2 = a^2,\ z = b(< 0)$ oriented counterclockwise.

**Solution.** By Stokes' theorem,

$$\iint_S (\nabla \times \mathbf{F}) \cdot \hat{n}\, dS = \oint_C \mathbf{F} \cdot d\mathbf{r}$$

Take $S$ as the plane circular disc $x^2 + y^2 \leq a^2$ in the plane $z = b$.
Then $\hat{n} = \mathbf{k}$.

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y & xz^2 & -zy^3 \end{vmatrix}$$

$$= -(3y^2z + 3xz^2)\mathbf{i} + (z^3 - 1)\mathbf{k}$$

$$(\nabla \times \mathbf{F}) \cdot \hat{n} = (b^3 - 1)$$

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (b^3 - 1)\, dS = (b^3 - 1)\pi a^2$$

**Example 4.15** Evaluate

$$\oint_C (e^x dx + 2y\, dy - dz)$$

using Stokes' theorem, where $C$ is the curve $x^2 + y^2 = 1,\ z = 1$ described clockwise as viewed from the origin.

**Solution.** By Stokes' theorem,

$$\iint_S (\nabla \times \mathbf{F}) \cdot \hat{n}\, dS = \oint_C \mathbf{F} \cdot d\mathbf{r}$$

Take $S$ as the circular disc $x^2 + y^2 \leq 1$ in the plane $z = 1$.
Then $\hat{n} = \mathbf{k}$.

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ e^x & 2y & -1 \end{vmatrix} = 0$$

$$\therefore \oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S 0\, dS = 0$$

## 4.7 Exercises

1. Evaluate

$$\iint_S p\left(\frac{x^4}{a^4} + \frac{y^4}{b^4} + \frac{z^4}{c^4}\right) dS,$$

where $S$ is the surface of the ellipsoid

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1,$$

and $p$ is the perpendicular from the origin to the tangent plane at $(x, y, z)$.

2. Using the divergence theorem, evaluate:

   (a) $\iint_S (yz\,dy\,dz + zx\,dz\,dx), \quad S : x^2 + y^2 + z^2 = 1$

   (b) $\iint_S e^{yz}\,dx\,dz, \quad S$ : surface of the parallelepiped $0 \le x \le 3,\ 0 \le y \le 2,\ 0 \le z \le 1$

   (c) $\iint_S (y^2z^2\mathbf{i} + x^2z^2\mathbf{j} + xy^2\mathbf{k}) \cdot \hat{n}\,dS$, $S$ : part of the sphere $x^2 + y^2 + z^2 = 1$ bounded by the $xy$-plane

3. Using the divergence theorem, evaluate $\iint_S \mathbf{F} \cdot \hat{n}\,dS$, where

   (a) $\mathbf{F} = x\mathbf{i} - y\mathbf{j} + (z^2 - 1)\mathbf{k}, \quad S$ : closed cylinder $x^2 + y^2 = 4,\ z = 0,\ z = 1$

   (b) $\mathbf{F} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}, \quad S$ : closed surface bounded by the cone $x^2 + y^2 = z^2$ and the plane $z = 1$

4. Using Stokes' theorem, evaluate

$$\iint_S [(y - z)\,dy\,dz + (z - x)\,dz\,dx + (x - y)\,dx\,dy],$$

   where $S$ is the portion of the surface

$$x^2 + y^2 - 2ax + az = 0, \quad z \ge 0.$$

5. Evaluate $\iint_S (\nabla \times \mathbf{F}) \cdot \hat{n}\,dS$,

$$\mathbf{F} = xy\mathbf{i} + yz\mathbf{j} + zx\mathbf{k}$$

   by Stokes' theorem over the surface of the unit cube $z = 1,\ y = n+1,\ x = m,\ x = m+1$, if the face $z = 0$ in the $xy$-plane is missing.

6. Using Stokes' theorem, evaluate

$$\iint_S (\nabla \times \mathbf{F}) \cdot \hat{n}\,dS, \quad \mathbf{F} = y\mathbf{i} + z\mathbf{j} + x\mathbf{k}$$

   over the surface $S : x^2 + y^2 + z^2 = 1,\ z \ge 0$.

7. Evaluate

$$\oint_C (y\,dz + x\,dx + z\,dy)$$

   by Stokes' theorem, where $C$ is the intersection of $x^2 + y^2 = 9$ and $z = y^2 + 1$, in the clockwise sense as viewed from origin.

8. Evaluate

$$\oint_C [(x + y)\,dx + (2x - z)\,dy + (y + 2z)\,dz]$$

   by Stokes' theorem, where $C$ is the boundary of the triangle with vertices $(2, 0, 0)$, $(0, 3, 0)$, $(0, 0, 6)$ in clockwise sense.

9. Apply Stokes' theorem to evaluate

$$\oint_C (y\,dx + z\,dy + x\,dz),$$

where $C$ is the intersection of $x^2 + y^2 + z^2 = a^2$ and $x + z = a$, in clockwise sense as viewed from the origin.

10. Use Stokes' theorem to prove that

$$\nabla \times (\nabla\phi) = 0,$$

where $\phi$ is any scalar function.